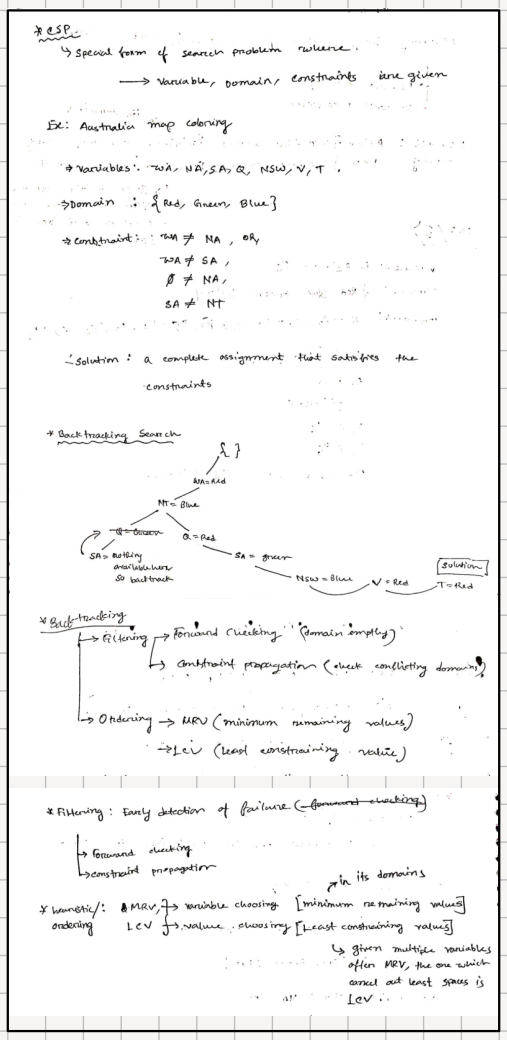
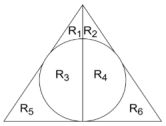
CSP(Constraint Satisfaction Problem)



1. Variables: Things we need to find values for.
   Ex> In a Sudoku puzzle each empty cell is a variable that needs a number.

2. Domains: The set of possible values that a variable can have.
   Ex>In Sudoku the domain for each cell is the numbers 1 to 9.

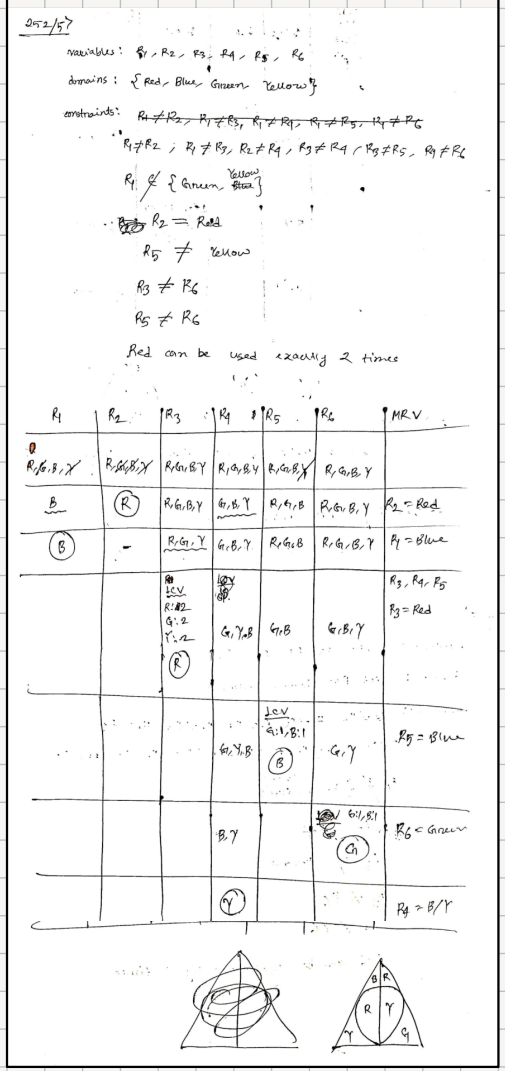
5. Consider the following "Deathly Hallows" symbol consisting of **6 regions**. You need to fill each region with one of the **4 colors: Red, Blue, Green, or Yellow**, in such a way that all 6 regions are distinguishable after assigning the colors.



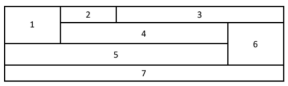
Some additional constraints:

- Region 1 cannot use Green or Yellow.
- Region 2 must be colored Red.
- Region 5 cannot use Yellow.
- Region 3 and Region 6 must not share the same color.
- Region 5 must be different from Region 6.
- Red can be used exactly 2 times.

Now formulate the problem as a **Constraint Satisfaction Problem (CSP)**. You must apply both **MRV and LCV** heuristics while choosing and assigning a color to a tile. Show the steps. **[5]**
[Hint: Do not assign same color to adjacent tiles]



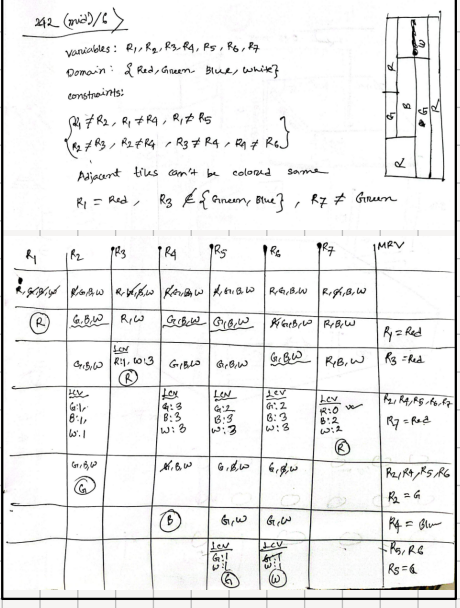
6. Consider the following 7 tiles. You need to fill each tile with one of the **four** colors: Red, Green, Blue, White in such a way that all the 7 tiles are distinguishable after assigning the colors.



However, there are **some additional constraints**:

- Tile 1 must be colored with **Red**
- Tile 3 **cannot** be colored with **Green** or **Blue**
- Tile 7 **cannot** be colored with **Green**

Now formulate the problem as a Constraint Satisfaction Problem (CSP). You **must apply both MRV** and **LCV heuristics** while choosing and assigning a color to a tile. **Show the steps**. **[5]**
[Hint: Do not assign same color to adjacent tiles]



Probability

(17) Probability

Up until this point, every topic's environment was deterministic.

From now on, there will uncertainty (probability)

Random Variables (aspect about which we have uncertainty):
- Variables denoted with [capital letter]
- Values denoted with [small letter]

Ex: R = is it raining? → {True, False} } Discrete Random Variables
T = is is hot or cold? → {hot, cold}
D = how long will it take to reach office? → [0, ∞) } Continuous Random Variables

Probability Distribution:
- Associate a probability with each value of random variable

Ex: (i) Temperature

P(T)	
T	P
hot	0.5
cold	0.5

(ii) Weather

P(W)	
W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

* $\forall_x P(X=x) \geq 0$ (for all 'x')
* $\sum_x P(X=x) = 1$ } $0 \leq P \leq 1$

Joint Distributions (probability distribution of multiple Random Var...)
- temperature, weather
- T = hot/cold
- W = sun/rain
- num of rows = 2×2 = 4

Ex: P(T, W)

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

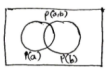
Events
- ⇒ P (hot AND sunny) = 0.4
- ⇒ P (hot) = 0.4 + 0.1 = 0.5
- ⇒ P (hot OR sunny) = 0.4 + 0.1 + 0.2 = 0.7

* A [probabilistic model] is a (joint distribution) over a set of random variables.

Marginal distributions
- Marginal distributions are sub-tables which eliminate variables
- Marginalization (summing out): Combine collapsed rows by adding

Conditional Probabilities:
- P(a | b) = P(a | given 'b')
- Probability of 'a' happening, given that 'b' has already happened.

* $P(a \mid b) = \frac{P(a,b)}{P(b)}$



Ex: P(T, W)

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

P(W = sun | T = cold) = ??
- cold is given

↳ $P(W=sun \mid T=cold) = \frac{P(W=sun, T=cold)}{P(T=cold)}$ (AND)

$= \frac{0.2}{0.2+0.3}$

$= \frac{0.2}{0.5}$

Ex: P(X, Y)

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

⇒ $P(+x \mid +y) = \frac{P(+x,+y)}{P(+y)} = \frac{0.2}{0.2+0.4}$

⇒ $P(-x \mid +y) = \frac{P(-x,+y)}{P(+y)} = \frac{0.4}{0.2+0.4}$

⇒ $P(-y \mid +x) = \frac{P(-y,+x)}{P(+x)} = \frac{0.3}{0.2+0.3}$

Conditional distribution

Ex: P(T, W)

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Joint Distribution

P(W | T = hot) (given)

W	P
sun	0.8
rain	0.2

conditional distribution

P(W | T = cold)

W	P
sun	0.4
rain	0.6

conditional distribution

P(W | T)

P(W = sun | T = cold) ⇒ $\frac{P(S,C)}{P(C)} \Rightarrow \frac{0.2}{0.2+0.3} \Rightarrow 0.4$

query, evidence variable

P(a | b)
- → P(W = sun | T = cold) = a value
- → P(W | T = cold) = a conditional distribution table
- → P(W | T) = a group of conditional distribution tables:
  - → Table 1 = P(W | T = hot)
  - → Table 2 = P(W | T = cold)

Full joint distribution

Ex: survey → Gender = male / female = 2
Category = student / teachers = 2
Juice = Orange / mango / lemon = 3
∴ rows = 2×2×3 = 12 rows

* Normalization = to make some entries sum equals '1'

(sum of entry / number of entries)

W	P
sun	0.2
rain	0.3

Normalize: $Z = 0.2+0.3 = 0.5$

W	P
sun	0.2/0.5 ⇒ 0.4
rain	0.3/0.5 ⇒ 0.6

Σ(1)

Probabilistic Inference
- calculate conditional probability from full joint distribution table.
- or
- compute a desired probability from known probabilities

* Inference by Enumeration

First? Identify
- Evidence variables ⇒ P(... | ...) evidence / evidence (given)
- Query variables ⇒ P(a | ..)
- Hidden variables ⇒ irrelevant for current question

Ex: Given Joint Distribution table

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Steps
- (S-1) selected entries consistent with evidence
- (S-2) Eliminate hidden variables to get joint of query and evidence
- (S-3) Normalize

(i) P(W | winter) ?
- Evidence → winter
- Query → weather
- Hidden → temp..

W	P
sun	0.25/0.5 = 0.5
rain	0.25/0.5 = 0.5

↳ Z = 0.5

(ii) P(W) ?
- Evidence → (not given)
- Query → W
- Hidden → T, S

W	P
sun	0.3+0.1+0.1+0.15 = 0.65
rain	0.05+0.05+0.05+0.2 = 0.35

(already normalized)

(iii) P(W | winter, hot) ?
- Evidence → winter, hot
- Query → W
- Hidden → X

W	P
sun	0.1/0.15 = 0.67
rain	0.05/0.15 = 0.33

Z = 0.1+0.05 = 0.15

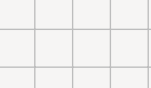
Full Joint Probability Distribution

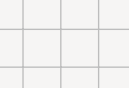
- Juice bar survey:
  - 8 male students, 5 female students, 3 male teachers and 2 female teachers bought orange juice
  - 6 male students, 7 female students, 2 male teachers and 4 female teachers bought mango juice
  - 4 male students, 5 female students, 2 male teachers and 2 female teachers bought lemon juice

Gender(G)	Category(C)	Juice(J)	P(G,C,J)
M	S	O	8/50
M	S	Mn	6/50
M	S	L	4/50
M	T	O	3/50
M	T	Mn	2/50
M	T	L	2/50
F	S	O	5/50
F	S	Mn	7/50
F	S	L	5/50
F	T	O	2/50
F	T	Mn	4/50
F	T	L	2/50

Inference by Enumeration

- General case:
  - Evidence variables: $E_1 \dots E_k = e_1 \dots e_k$
  - Query* variable: Q
  - Hidden variables: $H_1 \dots H_r$
  - $X_1, X_2, \dots X_n$ All variables
- We want: $P(Q | e_1 \dots e_k)$ * Works fine with multiple query variables, too
- Step 1: Select the entries consistent with the evidence
- Step 2: Sum out H to get joint of Query and evidence
- Step 3: Normalize $\times \frac{1}{Z}$





Ex: From Juice Survey table find P(Orange | Teachers)?

Soln: Evidence → Teachers [category]
Query → Orange [Juice]
Hidden → Gender

* we always make table depending on Query.

Oranges	Juice	P
	Orange	3/5+2/5 = 0.1/0.3 = 0.33
	Mango	2/5+4/5 = 0.4/0.3 = 0.4
	lemon	2/5+2/5 = 0.4/0.3 = 0.27

Z = 0.1+0.12+0.08 = 0.3

∴ P(O | T) = 0.33

The Product Rule:
- Sometimes we have conditional probability but we want the joint probability:

$\therefore P(x,y) = P(x \mid y) \cdot P(y)$ (joint, conditional)

$P(x \mid y) = \frac{P(x,y)}{P(y)}$ ↳ Bayes

The Chain Rule:
- Joint prob... from condition... of 2 variables = Product Rule
- Same thing for more than 2 var... = Chain Rule

* $P(x_1, x_2, x_3) = P(x_1) \cdot P(x_2 \mid x_1) \cdot P(x_3 \mid x_1, x_2)$
* $P(x_1, x_2, x_3, x_4) = P(x_1) \cdot P(x_2 \mid x_1) \cdot P(x_3 \mid x_1, x_2) \cdot P(x_4 \mid x_1, x_2, x_3)$

Bayes Rule:

* $P(x \mid y) = \frac{P(y \mid x)\, P(x)}{P(y)}$

Ex: M: meningitis, S: stiff neck
- P(+m) = 0.00001
- P(+s | +m) = 0.8
- P(+s | -m) = 0.01
- P(+m | +s) = ?

$\therefore P(+m \mid +s) = \frac{P(+s \mid +m)\, P(+m)}{P(+s)}$

$= \frac{P(+s \mid +m)\, P(+m)}{P(+s \mid +m) P(+m) + P(+s \mid -m) P(-m)}$

$= \frac{0.8 \times 0.0001}{(0.8 \times 0.0001) + (0.01 \times (1 - 0.00001))}$

$P(+s) = P(+s, +m) + P(+s, -m)$
$= P(+s \mid +m)\, P(+m) + P(+s \mid -m)\, P(-m)$

Ex)

Given

$P(W)$

R	P
sun	0.8
rain	0.2

$P(D|W)$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3

$\therefore P(W|dry) = ?$

Soln:

$$P(W|dry) = \frac{P(dry|W)\,P(W)}{P(dry)}$$

First) $P(sun|dry) = \frac{P(dry|sun)\,P(sun)}{P(dry)}$

$= \frac{0.9 \times 0.8}{0.78}$

$= 0.92$

$\Rightarrow P(dry)$

$\Rightarrow P(dry, sun) + P(dry, rain)$

$\Rightarrow P(d|s)\cdot P(s) + P(d|r)\cdot P(r)$

$\Rightarrow (0.9 \times 0.8) + (0.3 \times 0.2)$

$\Rightarrow 0.78$

So, $P(rain|dry) = 1 - 0.92 = 0.08$

$\therefore$

W	P
sun	0.92
rain	0.08

Ans

Probabilistic Independence:

↳ If two random variable doesn't influence each other

$\forall x, y : P(x, y) = P(x) \cdot P(y)$

or

$\forall x|y : P(x|y) = P(x)$

*we write as $(X \perp\!\!\!\perp Y) \Rightarrow X$ is independent of Y

* Doing checking, if one row found its not independent, other rows won't be independent.

** True independence is rarely in real-world.

↳ so conditional independence.

Conditional Independence: (2 var independent, grd var.. influence other 2)

$\forall x, y|z : P(x, y|z) = P(x|z) \cdot P(y|z)$

or

$\forall x, y, z : P(x|z, y) = P(x|z)$

$X \perp\!\!\!\perp Y | Z$ (Z: influencer)

↳ 'z' is the one influencing other two.

- P(Toothache, Cavity, Catch)
- If I have a cavity, the probability that the probe catches in it doesn't depend on whether I have a toothache:
  - P(+catch | +toothache, +cavity) = P(+catch | +cavity)
- The same independence holds if I don't have a cavity:
  - P(+catch | +toothache, -cavity) = P(+catch| -cavity)
- Catch is *conditionally independent* of Toothache given Cavity:
  - P(Catch | Toothache, Cavity) = P(Catch | Cavity)
- Equivalent statements:
  - P(Toothache | Catch , Cavity) = P(Toothache | Cavity)
  - P(Toothache, Catch | Cavity) = P(Toothache | Cavity) P(Catch | Cavity)
  - One can be derived from the other easily

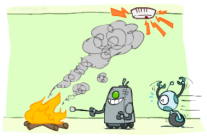
Conditional Independence and chain Rule

original) $P(Rain, Traffic, Umbrella) = P(Rain) \cdot P(Traffic|Rain) \cdot P(Umbrella|Rain, Traffic)$

with assumption of conditional independence

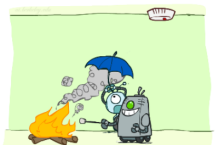
$P(T, R, U) = P(R) \cdot P(T|R) \cdot P(U|R)$ ($[T \perp\!\!\!\perp U | R]$: R influences)

- What about this domain:
  - Fire
  - Smoke → Z
  - Alarm



$\Rightarrow (F \perp\!\!\!\perp A | S)$

↳ fire is independent of alarm, given smoke.



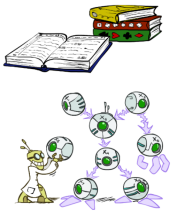
- What about this domain:
  - Traffic
  - Umbrella
  - Raining → Z

So, $(T \perp\!\!\!\perp U | R)$

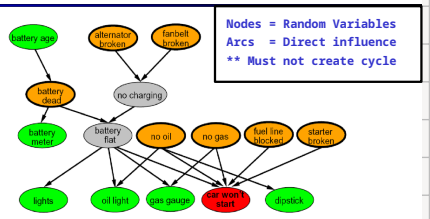
Bayes' Nets

Bayes' Nets: Big Picture

- Two problems with using full joint distribution tables as our probabilistic models:
  - Unless there are only a few variables, the joint is WAY too big to represent explicitly
  - Hard to learn (estimate) anything empirically about more than a few variables at a time
- Bayes' nets: a technique for describing complex joint distributions (models) using simple, local distributions (conditional probabilities)
  - More properly called graphical models
  - We describe how variables locally interact
  - Local interactions chain together to give global, indirect interactions
  - For about 10 min, we'll be vague about how these interactions are specified



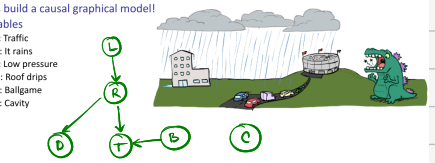
Example Bayes' Net: Car



Nodes = Random Variables
Arcs = Direct influence
** Must not create cycle

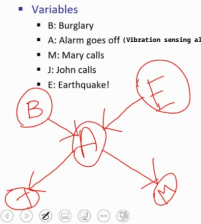
Example: Traffic II

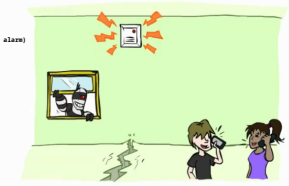
- Let's build a causal graphical model!
- Variables
  - T: Traffic
  - R: It rains
  - L: Low pressure
  - D: Roof drips
  - B: Ballgame
  - C: Cavity



Example: Alarm Network

- Variables
  - B: Burglary
  - A: Alarm goes off (Vibration sensing alarm)
  - M: Mary calls
  - J: John calls
  - E: Earthquake!



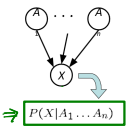


Bayes' Net Semantics

- A set of nodes, one per variable X
- A directed, acyclic graph
- A conditional distribution for each node
  - A collection of distributions over X, one for each combination of parents' values

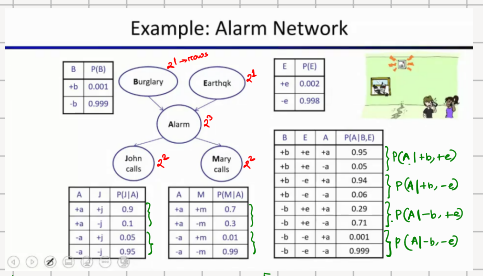
$$P(X|a_1 \dots a_n)$$

  - CPT: conditional probability table
  - Description of a noisy "causal" process



A Bayes net = Topology (graph) + Local Conditional Probabilities

Example: Alarm Network



If we built this using full joint → $2^5 = 32$ rows; here = total 20 rows

Inference: calculating some useful quantity from a joint probability distribution

Probabilities in BNs

- Why are we guaranteed that setting

$$P(x_1, x_2, \dots x_n) = \prod_{i=1}^{n} P(x_i|parents(X_i))$$

results in a proper joint distribution?

- Chain rule (valid for all distributions):

$$P(x_1, x_2, \dots x_n) = \prod_{i=1}^{n} P(x_i|x_1 \dots x_{i-1})$$

- Assume conditional independences:

$$P(x_i|x_1, \dots x_{i-1}) = P(x_i|parents(X_i))$$

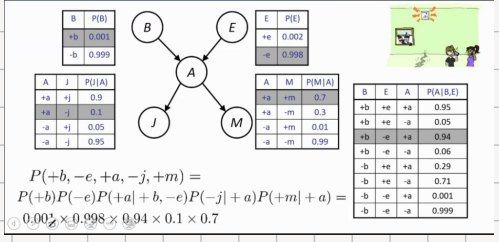
→ Consequence:

$$P(x_1, x_2, \dots x_n) = \prod_{i=1}^{n} P(x_i|parents(X_i))$$

- Not every BN can represent every joint distribution
  - The topology enforces certain conditional independencies

Calculate Joint Probability from Bayes' Net

Example: Alarm Network



B	P(B)
+b	0.001
-b	0.999

E	P(E)
+e	0.002
-e	0.998

A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

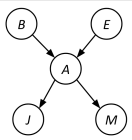
$$P(+b, -e, +a, -j, +m) = P(+b)P(-e)P(+a|+b,-e)P(-j|+a)P(+m|+a) = 0.001 \times 0.998 \times 0.94 \times 0.1 \times 0.7$$

Inference by Enumeration in Bayes' Net

- Given unlimited time, inference in BNs is easy
- Reminder of inference by enumeration by example:

$$P(B|+j,+m)$$

P(+b | +j,+m)
P(-b | +j,+m)



For Hidden Variables:
A E
+a +e
+a -e
-a +e
-a -e

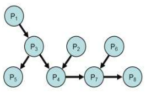
Net has 5 variables, but we are told to find P( ) between 3. So there must be 2 hidden variables that must be taken into consideration
Hidden: [A, E]

Independence in Bayes Nets

Every variable in a bayes net is conditionally independent of its non-descendents given its parents.

Determine if the following statements are True/False:

i. Pr(P8|P3,P7)=Pr(P8|P7)
ii. Pr(P3|P1,P5)=Pr(P3|P1)
iii. Pr(P4,P2|P3)=Pr(P4|P3)Pr(P2|P3)
iv. Pr(P5,P4|P1)=Pr(P5|P1)Pr(P4|P1)



1. (P8|P3 | P7) ==> ?
2. (P3|P1 | P5) ==> ? [P3 and P5 has parent-child relationship]
3. (P4,P2 | P3) ==> ? [P4 and P2 has parent-child relationship]
4. (P5|P4 | P1) ==> ? [P5 and P4 are independent, but P1 is neither's parent]

Inference by Enumeration in Bayes' Net

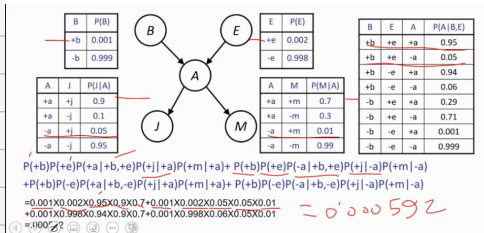
P(+b|+j,+m)=P(+b,+j,+m)/P(+j,+m)

$[P(X|\bar{a}) = \frac{P(X,\bar{a})}{P(\bar{a})}]$

P(+b,+j,+m) [now converting 3→5 variables]

=P(+b,+e,+a,+j,+m)+P(+b,+e,-a,+j,+m)+P(+b,-e,+a,+j,+m)+P(+b,-e,-a,+j,+m)

=P(+b)P(+e)P(+a|+b,+e)P(+j|+a)P(+m|+a)+ P(+b)P(+e)P(-a|+b,+e)P(+j|-a)P(+m|-a)
+P(+b)P(-e)P(+a|+b,-e)P(+j|+a)P(+m|+a)+ P(+b)P(-e)P(-a|+b,-e)P(+j|-a)P(+m|-a)



P(+b)P(+e)P(+a|+b,+e)P(+j|+a)P(+m|+a)+ P(+b)P(+e)P(-a|+b,+e)P(+j|-a)P(+m|-a)
+P(+b)P(-e)P(+a|+b,-e)P(+j|+a)P(+m|+a)+ P(+b)P(-e)P(-a|+b,-e)P(+j|-a)P(+m|-a)
=0.001X0.002X0.95X0.9X0.7+0.001X0.002X0.05X0.05X0.01
+0.001X0.998X0.94X0.9X0.7+0.001X0.998X0.06X0.05X0.01
= 0.000592

```
P(+j,+m) = P(+b,+j,+m) + P(-b,+j,+m)
         = 0.000592    + P(-b,+j,+m)

P(-b,+j,+m)
= P(-b,+e,+a,+j,+m) + P(-b,+e,-a,+j,+m) +
  P(-b,-e,+a,+j,+m) + P(-b,-e,-a,+j,+m)

= P(-b) P(+e) P(+a|-b,+e) P(+j|+a) P(+m|+a) +
  P(-b) P(+e) P(-a|-b,+e) P(+j|-a) P(+m|-a) +
  P(-b) P(-e) P(+a|-b,-e) P(+j|+a) P(+m|+a) +
  P(-b) P(-e) P(-a|-b,-e) P(+j|-a) P(+m|-a)
= 0.0014917

So, P(+j,+m) = P(+b,+j,+m) + P(-b,+j,+m)
             = 0.000592    + 0.0014917
             = 0.0020837


Hence: P(+b|+j,+m) = P(+b,+j,+m)/P(+j,+m)
                   = 0.000592 / 0.0020837
                   = 0.284
Thus:  P(-b|+j,+m) = 1 - (0.284)
                   = 0.716
```

∴ P(B|+j,+m) ⇒

B	P
+b	0.284
-b	0.716

Naïve Bayes Classifier

Naive Bayes calculates the probability of each possible class given the observed features, then chooses the class with the highest probability.

label & features represented with random var(nodes)

Other Classification Tasks

- Classification: given inputs x, predict labels (cl
- Examples:
  - Spam detection (input: document, classes: spam / ham)
  - OCR (input: images, classes: characters)
  - Medical diagnosis (input: symptoms, classes: diseases)
  - Automatic essay grading (input: document, classes: grades)
  - Fraud detection (input: account activity, classes: fraud / no fraud)
  - Customer service email routing
  - ... many more
- Classification is an important commercial technology!

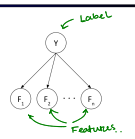
Model-Based Classification

- Model-based approach
  - Build a model (e.g. Bayes' net) where both the label and features are random variables
  - Instantiate any observed features
  - Query for the distribution of the label conditioned on the features
- Challenges
  - What structure should the BN have?
  - How should we learn its parameters?

General Naïve Bayes

- A general Naive Bayes model:
  Features are conditionally independent of each other given the class labels (Assumption)

$$P(Y, F_1 \dots F_n) = \underbrace{P(Y)}_{\text{table for class label}} \prod_i \underbrace{P(F_i|Y)}_{\text{table for features}}$$

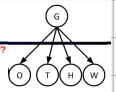


P(Y, F1, F2, … Fn) = P(Y) * P(F1 | Y) * P(F2|Y) * … * P(Fn|Y)

- We only have to specify how each feature depends on the class
- Total number of parameters is *linear* in n
- Model is very simplistic, but often works anyway

Example

Today is Sunny, Hot, Humidity Normal and No wind. What is probability of playing golf?



Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

Naive Bayes Classifier Steps:
(S-1) Trainig Phase
>> Populate tables for every node

(S-2) Testing Phase
>> Given some features, predict the class

(S-1) Training Phase

P(G)

G	P
yes	9/14
no	5/14

P(W|G)

W	G	P
True	yes	3/9
True	no	3/5
False	yes	6/9
False	no	2/5

P(H|G)

H	G	P
high	yes	3/9
high	no	4/5
normal	yes	6/9
normal	no	1/5

P(O|G)

O	G	P
Rainy	yes	2/9
Overcast	no	0/5
Sunny	yes	3/9
Rainy	no	3/5
Overcast	yes	4/9
Sunny	no	2/5

P(T|G)

T	G	P
hot	yes	2/9
mild	no	2/5
cool	yes	3/9
hot	no	2/5
mild	yes	4/9
cool	no	1/5

(S-2) Testing Phase

Today is Sunny, Hot, Humidity Normal and No wind. What is probability of playing golf? features ⇒ ⟨sunny, hot, normal, false⟩

So, $P(G \mid \text{sunny, hot, normal, false})$

$P(\text{yes} \mid \text{sunny, hot, normal, false}) = \frac{P(\text{yes} \mid \text{sunny, hot, normal, false})}{P(\text{sunny, hot, normal, false})}$

$P(\text{no} \mid \text{sunny, hot, normal, false}) = \frac{P(\text{no} \mid \text{sunny, hot, normal, false})}{P(\text{sunny, hot, normal, false})}$

This part is common. So, instead of calculation find proportionality

$P(\text{yes} \mid \text{sunny, hot, normal, false}) \propto P(y, s, h, n, f)$

$\propto P(y) \cdot P(s|y) \cdot P(h|y) \cdot P(n|y) \cdot P(f|y)$

$\propto 9/14 \times 3/9 \times 2/9 \times 6/9 \times 6/9$

$\propto 0.0212$

$P(\text{no} \mid \text{sunny, hot, normal, false}) \propto P(n, s, h, n, f)$

$\propto P(n) \cdot P(s|n) \cdot P(h|n) \cdot P(n|no) \cdot P(f|n)$

$\propto 5/14 \times 3/5 \times 4/5 \times 1/5 \times 2/5$

$\propto 0.0046$

here, our result is not normalized >

So, $P(\text{yes} \mid \text{features}) + P(\text{no} \mid \text{features}) = 0.0212 + 0.0046$

$= 0.0258$

normalization >

$P(\text{yes} \mid \text{features}) = \frac{0.0212}{0.0258} \times 100 = 82.17\%$

$P(\text{no} \mid \text{features}) = \frac{0.0046}{0.0258} \times 100 = 17.83\%$

Yes, this day is suitable for playing golf

Drawback of Naive Bayes Classifier:

>> If any probability value becomes '0', the whole product will be '0'

>> To avoid this, we use smoothing technique(Laplace smoothing)

- Laplace's estimate (extended):
  - Pretend you saw every outcome k extra times

$$P_{LAP,k}(x) = \frac{c(x) + k}{N + k|X|}$$

  - What's Laplace with k = 0?
  - k is the strength of the prior
- Laplace for conditionals:
  - Smooth each condition independently:

$$P_{LAP,k}(x|y) = \frac{c(x, y) + k}{c(y) + k|X|}$$

Ex> $P(\text{hot} \mid \text{yes}) = \frac{2}{9}$

K=2> $P(\text{hot} \mid \text{yes}), k=2 = \frac{2+2}{9+2\times3}$ (features)

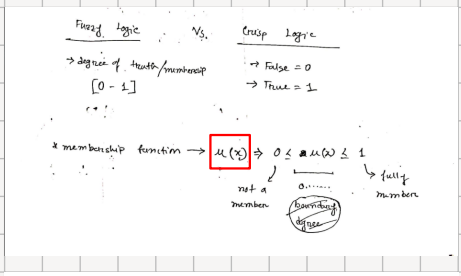
K=1> $P(\text{hot} \mid \text{yes}), k=1 = \frac{2+1}{9+1\times3}$ (features)

ex> $P(\text{overcast} \mid \text{no}) = \frac{0}{5}$

$P_{LAP,K=1} = \frac{0+1}{5+(1\times3)}$ overcast, sunny, rainy

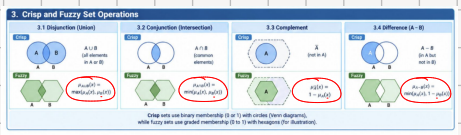
$P_{LAP,K=3} = \frac{0+3}{5+(3\times3)}$

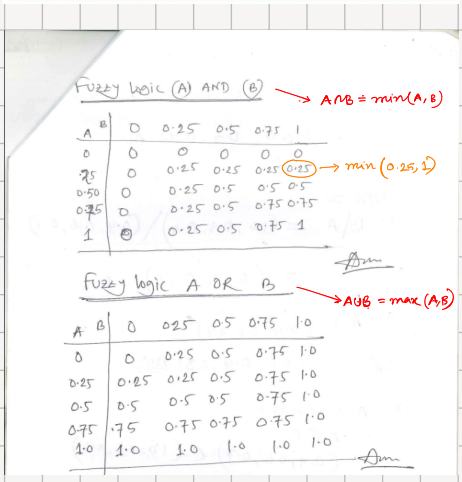
Fuzzy & Crisp Logic

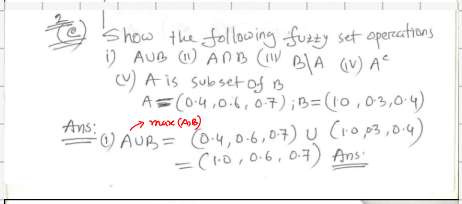


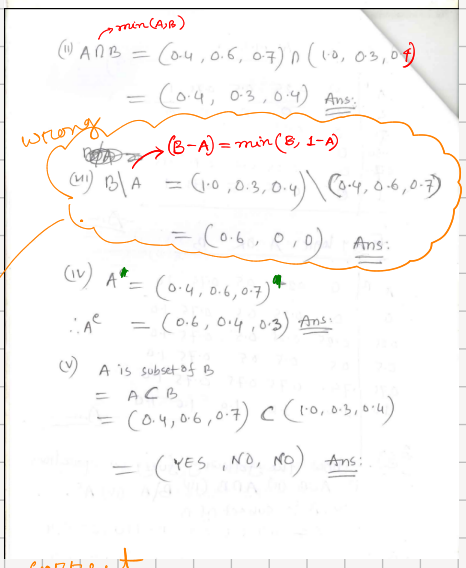
Fuzzy Logic Vs Probability

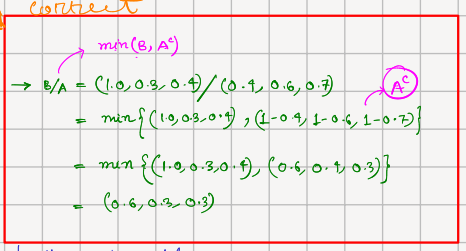
- Both operate over the same numeric range and at first glance both have similar values:0.0 representing false(or non-membership) and 1.0 representing true.
- In terms of probability: the natural language statement would be "there is an 80% chance that Jane is old."
- While the fuzzy terminology corresponds to "Jane's degree of membership within the set of old people is 0.80."









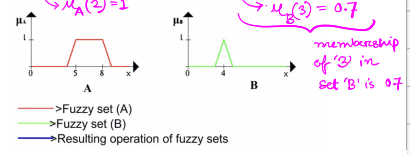


Operations on Fuzzy Set

Consider:

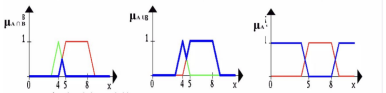
A= {1/2 + .5/3 + .3/4 + .2/5} → $\mu_A(2) = 1$

B= {.5/2 + .7/3 + .2/4 + .4/5} → $\mu_B(3) = 0.7$ membership of '2' in set 'B' is 0.7



>Fuzzy set (A)
>Fuzzy set (B)
>Resulting operation of fuzzy sets

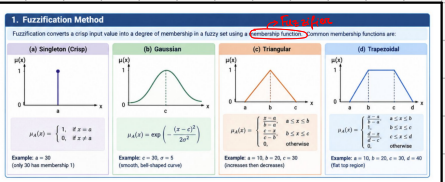
INTERSECTION (A ^ B)	UNION (A v B)	COMPLEMENT (¬A)



$\mu_{A\cap B} = \min(\mu_A(x), \mu_B(x))$
{.5/2 + .5/3 + .2/4 + .2/5}

$\mu_{A\cup B} = \max(\mu_A(x), \mu_B(x))$
{1/2 + .7/3 + .3/4 + .4/5}

$\mu_{A'} = 1 - \mu_A(x)$
{1/1 + 0/2 + .5/3 + .7/4 + .8/5} not necessary



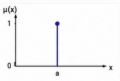
1. Singleton fuzzifier — easiest

Suppose the input is:

$$x = 30$$

A singleton fuzzifier is:

$$\mu(x) = \begin{cases} 1 & x = 30 \\ 0 & x \neq 30 \end{cases}$$



So if asked for membership at:

- $x = 30 \rightarrow 1$
- $x = 29 \rightarrow 0$
- $x = 31 \rightarrow 0$

There's basically no difficult math.

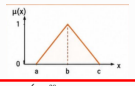
2. Triangular fuzzifier — very exam-friendly

Suppose:

$$A = (20, 30, 40)$$

This means:

- 20 → membership 0
- 30 → membership 1
- 40 → membership 0



The membership function is:

$$\mu_A(x) = \begin{cases} \frac{x-20}{30-20}, & 20 \le x \le 30 \\ \frac{40-x}{40-30}, & 30 \le x \le 40 \\ 0, & \text{otherwise} \end{cases}$$

Example: Find membership of $x = 25$

Since 25 is on the **left side**:

$$\mu_A(25) = \frac{25-20}{30-20}$$

$$= \frac{5}{10}$$

$$\boxed{0.5}$$

Example: $x = 35$

Now we're on the right side:

$$\mu_A(35) = \frac{40-35}{40-30}$$

$$= \frac{5}{10}$$

$$\boxed{0.5}$$

3. Trapezoidal fuzzifier

Suppose:

$$A = (20, 25, 35, 40)$$

The shape is:

The membership function is:

$$\mu_A(x) = \begin{cases} 0 & x < 20 \\ \frac{x-20}{25-20} & 20 \le x \le 25 \\ 1 & 25 \le x \le 35 \\ \frac{40-x}{40-35} & 35 \le x \le 40 \\ 0 & x > 40 \end{cases}$$

Example: $x = 23$

$$\mu_A(23) = \frac{23-20}{25-20}$$

$$= \frac{3}{5}$$

$$\boxed{0.6}$$

Example: $x = 30$

30 is inside the flat top:

$$\boxed{\mu_A(30) = 1}$$

Example: $x = 38$

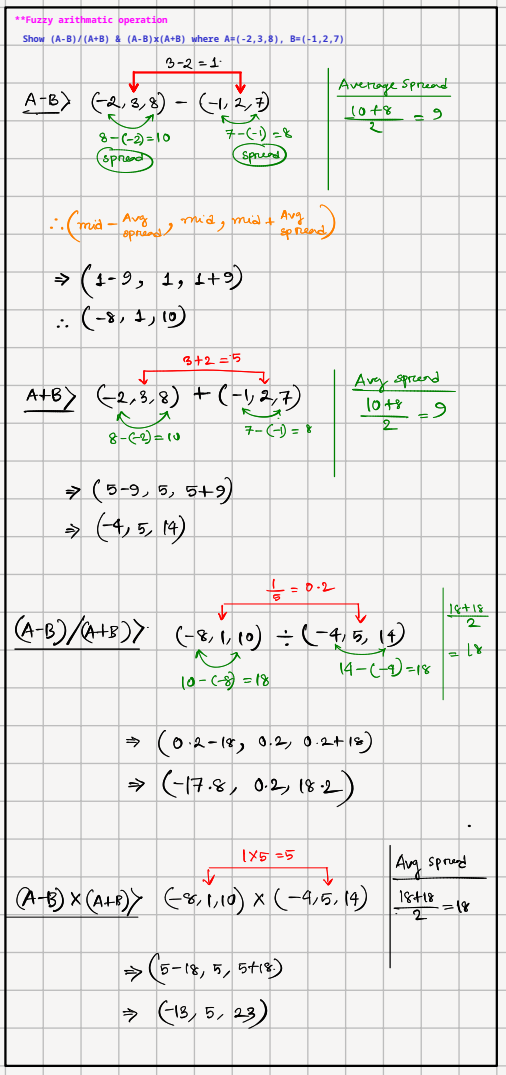
$$\mu_A(38) = \frac{40-38}{40-35}$$

$$= \frac{2}{5}$$

$$\boxed{0.4}$$

**Fuzzy arithmatic operation

Show (A-B)/(A+B) & (A-B)x(A+B) where A=(-2,3,8), B=(-1,2,7)

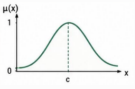


4. Gaussian fuzzifier

This one has a formula.

A Gaussian membership function is usually:

$$\mu_A(x) = e^{-\frac{(x-c)^2}{2\sigma^2}}$$



where:

- c = center
- σ = spread/width

Suppose:

$$c = 30, \quad \sigma = 5$$

and $x = 35$.

Then:

$$\mu_A(35) = e^{-\frac{(35-30)^2}{2(5)^2}}$$

$$= e^{-\frac{25}{50}}$$

$$= e^{-0.5}$$

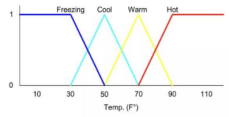
$$\boxed{\approx 0.607}$$

So 35°C has about **0.607 membership** in this fuzzy concept.

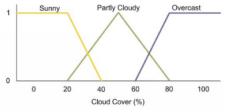
Fuzzification:
Calculate Input Membership Levels

Input:

Temp: {Freezing, Cool, Warm, Hot}



Cover: {Sunny, Partly cloudy, Overcast}



Rules

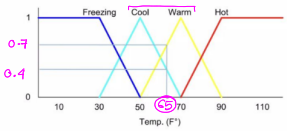
- If it's Sunny and Warm, drive Fast
  Sunny(Cover)∧Warm(Temp)⇒ Fast(Speed)
- If it's Cloudy and Cool, drive Slow
  Cloudy(Cover)∧Cool(Temp)⇒ Slow(Speed)
- Driving Speed is the combination of output of these rules...

Temp. (F°)

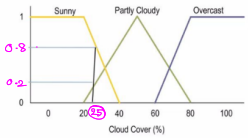
Example Speed Calculation

How fast will I go if it is
- 65 F
- 25 % Cloud Cover ?

- **65 F ⇒ Cool = 0.4, Warm= 0.7**



- **25% Cover ⇒Sunny = 0.8, Cloudy = 0.2**

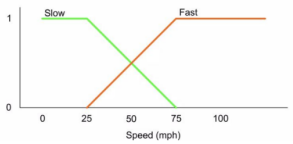


Calculating: (According to rules)

- If it's Sunny and Warm, drive Fast
  Sunny(Cover)∧Warm(Temp)⇒Fast(Speed)
  0.8 ∧ 0.7 = 0.7
  ⇒ **Fast = 0.7**
- If it's Cloudy and Cool, drive Slow
  Cloudy(Cover)∧Cool(Temp)⇒Slow(Speed)
  0.2 ∧ 0.4 = 0.2
  ⇒ **Slow = 0.2**

Output:

Speed: {Slow, Fast}



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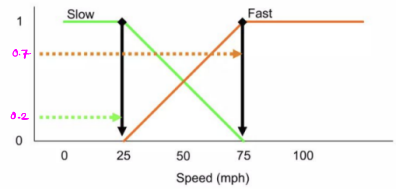
Defuzzification:
Constructing the Output

Defuzzification:
Constructing the Output

- Speed is 20% Slow and 70% Fast



- Find centroids: Location where membership is 100%
- Speed = weighted mean
  = (2*25+7*75)/(9)
  = 63.8 mph

Speed = $\frac{(0.2\times25)+(0.7\times75)}{0.7+0.2}$
= 63.8

Step 1: Identify the Centroids (Representative Values)

Find the value on the x-axis (Speed) where each fuzzy set has full (100% or 1.0) membership: PNG

- **Slow (Green line):** Reaches maximum membership (1.0) at $x = 25$ mph. PNG
- **Fast (Orange line):** Reaches maximum membership (1.0) at $x = 75$ mph. PNG

Step 2: Determine the Membership Weights

Extract the degree of membership (activation level) for each set given in the problem: PNG

- **Slow weight (w_1):** $20\% = 0.2$ PNG
- **Fast weight (w_2):** $70\% = 0.7$ PNG

(Note: In the slide's calculation, 0.2 and 0.7 are scaled by 10 to use whole numbers 2 and 7, which gives the exact same mathematical ratio). PNG

Step 3: Apply the Weighted Average Formula

The weighted mean formula is:

$$\text{Speed} = \frac{\sum(w_i \cdot x_i)}{\sum w_i} = \frac{(w_{\text{Slow}} \cdot x_{\text{Slow}}) + (w_{\text{Fast}} \cdot x_{\text{Fast}})}{w_{\text{Slow}} + w_{\text{Fast}}}$$

Substitute the values into the formula: PNG

1. **Numerator (Weighted Sum of Centroids):**

$$(0.2 \times 25) + (0.7 \times 75) = 5 + 52.5 = 57.5$$

(Or using scaled integer weights: $(2 \times 25) + (7 \times 75) = 50 + 525 = 575$) PNG

2. **Denominator (Sum of Weights):**

$$0.2 + 0.7 = 0.9$$

(Or using scaled integer weights: $2 + 7 = 9$) PNG

3. **Final Division:**

$$\text{Speed} = \frac{57.5}{0.9} = \frac{575}{9} \approx 63.888\ldots \text{ mph}$$

First Order Logic

FOL is a representation for AI model/agents.
With this representation AI can [Fact -> Conclusion]
FOL provides AI model with a precise way to represent objects, properties, relationships, and rules, enabling the system to reason logically.

Core Components>>
⚠Constants: Specific names or things (like John or NewYork).

⚠Variables: Placeholders for any object (like x or y).

⚠Predicates: Actions, traits, or relations like - [Brother(John, Bob)].

⚠Functions: Mappings from one object to another like - [FatherOf(John)].

⚠Quantifiers: Rules defining object scope, such as Universal (∀, meaning "for all") and Existential (∃, meaning "there exists").

⚠Logical Connectives: Include conjunction(∧), disjunction (∨), implication (→), biconditional(↔) and negation (¬).

Example of Reasoning
⚠Fact 1: ∀x(Human(x) → Mortal(x)) — "All humans are mortal."
⚠Fact 2: Human (Socrates) — "Socrates is a human."
⚠Inference: Mortal (Socrates) — "Socrates is mortal."

We have to convert this Inference in FOL so the machine understands it

Problem 1:
Suppose we know
  Fact: Socrates is a human ==> Human(Socrates)
  Rule: All humans are mortal ==> ∀x [Human(x)→ Mortal(x)]
From these AI can infer: Mortal(Socrates)

Problem 2: Students and courses
⚠Rahim is a student.
⚠Every student studies at least one course.
⚠Therefore, Rahim studies a course.
FOL representation:
 Fact: Student (Rahim)
 Rule: ∀x [Student(x)→ ∃y Studies(x,y)]
 Therefore: Student(Rahim)→∃y Studies(Rahim,y)
 Since: Student(Rahim)=True
 We conclude: ∃y Studies(Rahim, y)
 Meaning: There exists at least one course that Rahim studies.

Problem 3: Family relationship
Suppose: Everyone's father is a parent.
FOL: ∀x∀y [Father(y,x)→Parent(y,x)]
Given: Father(Ali,Rahim)
Then: Father(Ali,Rahim)→Parent(Ali,Rahim)
Therefore: Parent(Ali,Rahim)

Problem 4: Using AND (∧)
Suppose an AI system knows: If a person is a student and studies AI, then the person is an AI student.
FOL: ∀x[(Student(x)∧Studies(x,AI))→AIStudent(x)]
Given: Student(Tanvir) and Studies(Tanvir, AI)
Therefore: Student(Tanvir)∧Studies(Tanvir,AI) is True,
so: AIStudent(Tanvir)

Problem 5: Using NOT (¬)
Suppose: No student is a professor.
FOL: ∀x[Student(x)→¬Professor(x)]
Given: Student (Rahim)
Therefore: ¬Professor(Rahim)
So the AI can conclude that Rahim is not a professor.

Problem 6: A typical AI reasoning problem
Knowledge Base: Human(Ali), Human(Rahim)
Rule: ∀x[Human(x)→Mortal(x)]
Question: Is Ali mortal?
Substitute x=Ali
Human(Ali)→Mortal(Ali)
Since: Human(Ali)=True
We infer: Mortal(Ali)
Similarly, because:
Human(Rahim)=True
We infer: Mortal(Rahim)

**Important FOL pattern for exams
A very common structure is: ∀x(P(x)→Q(x))
Given: P(a) Infer: Q(a)
For example: ∀x(Student(x)→Intelligent(x))
Student(Ali)
Therefore: Intelligent(Ali)

This is essentially Universal Instantiation + Modus Ponens and it is one of the most important reasoning patterns in FOL for AI.

Sentences to Well formed Formulas in Predicate Logic

1. Marcus was a man.	1. $man(Marcus)$
2. Marcus was a Pompeian.	2. $Pompeian(Marcus)$
3. All Pompeian were Romans.	3. $\forall x : Pompeian(x) \rightarrow Roman(x)$
4. Caesar was a ruler.	4. $ruler(Caesar)$
5. All Romans were either loyal to Caesar or hated him.	5. $\forall x : Roman(x) \rightarrow loyalto(x, Caesar) \vee hate(x, Caesar)$
6. Everyone is loyal to someone.	6. $\forall x : \exists y : loyalto(x, y)$
7. People only try to assassinate rulers they are not loyal to.	7. $\forall x : \forall y : person(x) \wedge ruler(y) \wedge tryassassinate(x, y) \rightarrow \neg loyalto(x, y)$
8. Marcus tried to assassinate Caesar.	8. $tryassassinate(Marcus, Caesar)$
9. All men are people	9. $\forall x : man(x) \rightarrow person(x)$

Resolution using Predicate Logic in Artificial Intelligence

1. Was Marcus Loyal to Caesar?

nil
$\downarrow(1)$
$man(Marcus)$
$\downarrow (9)$
$person(Marcus)$
$\downarrow (8)$
$person(Marcus) \wedge tryassassinate(Marcus, Caesar)$
$\downarrow (4)$
$person(Marcus) \wedge tryassassinate(Marcus, Caesar) \wedge ruler(Caesar)$
$\downarrow$ (7, substitution)
$\neg loyalto(Marcus, Caesar)$

1. $man(Marcus)$
2. $Pompeian(Marcus)$
3. $\forall x : Pompeian(x) \rightarrow Roman(x)$
4. $ruler(Caesar)$
5. $\forall x : Roman(x) \rightarrow loyalto(x, Caesar) \vee hate(x, Caesar)$
6. $\forall x : \exists y : loyalto(x, y)$
7. $\forall x : \forall y : person(x) \wedge ruler(y) \wedge tryassassinate(x, y) \rightarrow \neg loyalto(x, y)$
8. $tryassassinate(Marcus, Caesar)$
9. $\forall x : man(x) \rightarrow person(x)$

Resolution using Predicate Logic in Artificial Intelligence

2. Was Marcus hates Caesar?

$\neg loyalto(Marcus, Caesar)$
$\downarrow (2)$
$Pompeian(Marcus)$
$\neg loyalto(Marcus, Caesar)$
$\downarrow (3)$
$Roman(Marcus)$
$\neg loyalto(Marcus, Caesar)$
$\downarrow (5)$
$hate(Marcus, Caesar)$

1. $man(Marcus)$
2. $Pompeian(Marcus)$
3. $\forall x : Pompeian(x) \rightarrow Roman(x)$
4. $ruler(Caesar)$
5. $\forall x : Roman(x) \rightarrow loyalto(x, Caesar) \vee hate(x, Caesar)$
6. $\forall x : \exists y : loyalto(x, y)$
7. $\forall x : \forall y : person(x) \wedge ruler(y) \wedge tryassassinate(x, y) \rightarrow \neg loyalto(x, y)$
8. $tryassassinate(Marcus, Caesar)$
9. $\forall x : man(x) \rightarrow person(x)$

https://www.youtube.com/watch?v=GgIe_tNpLjk

Resolution using Predicate Logic in Artificial Intelligence

1. **Marcus was a man.**
2. **Marcus was a Pompeian.**
3. **Marcus was born in 40 A.D.**
4. **All men are mortal.**
5. **All Pompeians died when the volcano erupted in 79 A.D.**
6. **No mortal lives longer than 150 years.**
7. **It is now 1991.**
8. **Alive means not dead.**
9. **If someone dies, then he is dead at all later times.**

Was Marcus Dead?

Was Marcus dead?

1. $man(Marcus)$
2. $Pompeian(Marcus)$
3. $born(Marcus, 40)$
4. $\forall x : man(x) \rightarrow mortal(x)$
5. $\forall : Pompeian(x) \rightarrow died(x, 79)$
6. $erupted(volcano, 79)$
7. $\forall_x : \forall t_1 : \forall t_2 : mortal(x) \wedge born(x, t_1) \wedge gt(t_2 - t_1, 150) \rightarrow dead(x, t_2)$
8. $now = 1991$
9. $\forall x : \forall t : [alive(x, t) \rightarrow \neg dead(x, t)] \wedge [\neg dead(x, t) \rightarrow alive(x, t)]$
10. $\forall x : \forall t_1 : \forall t_2 : died(x, t_1) \wedge gt(t_2, t_1) \rightarrow dead(x, t_2)$

nil
$\downarrow$ (compute gt)
$gt(1991, 79)$
$\downarrow$ (8, substitute equals)
$gt(now, 79)$
$\downarrow$ (2)
$Pompeian(Marcus) \wedge gt(now, 79)$
$\downarrow$ (5, substitution)
$died(Marcus, t_1) \wedge gt(now, t_1)$
$\downarrow$ (10, substitution)
$dead(Marcus, now)$
$\downarrow$ (9, substitution)
$\neg alive(Marcus, now)$

Machine Learning

Types>>
Supervised learning: Training data includes desired outputs
Unsupervised learning: Training data does not include desired outputs

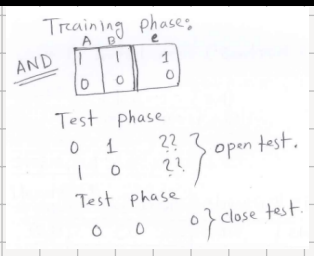
What do you understand by Machine Learning?
==> Machine Learngin is a branch of Artificial Intelligence. It is a scientific discipline with the design and development of algorithms that recognize complex patterns and make intelligent decisions based on input data.

Generalization: Generalization is the ability of an algorithm to perform accurately on new, unseen data after having trained on a learning data set.

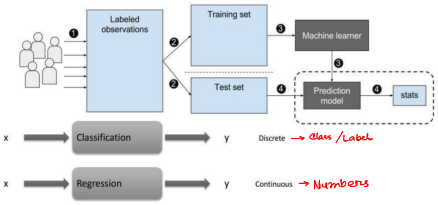
Data mining: Discovery of unknown properties on the data.

Closed test Vs Open Test

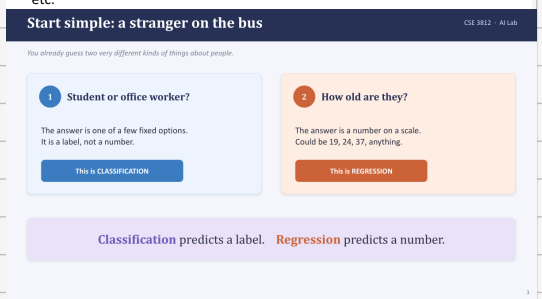
	Closed test	Open test
Test data	Known	Unknown/unseen
Input known?	✅	❌ Previously unseen
Output known beforehand?	✅	❌
Main idea	Test with known cases	Test with new cases



Supervised Learning



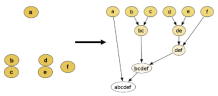
- Common Learning Algorithms: Linear Regression, Logistic Regression, Decision Tree, Random Forest, k-Nearest Neighbour, SVM, Neural Network etc.

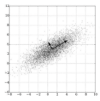


Given a set of data points {x1 ,x(2), …, x(m)} associated to a set of outcomes, {y1 ,y(2), …, y(m)}, we want to build a classifier that learns how to predict y from x.

Unsupervised Learning

- Looks for previously undetected patterns in a data set with no pre-existing labels.
  - Actual output y is absent!





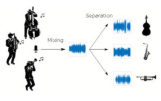


Fig: Agglomerative clustering Fig: PCA

Fig: ICA

- Common Learning Algorithms: K-means, hierarchical clustering, mixture model, Local outlier factor, EM, PCA, ICA, etc.

Regression(Supervised Learning)

Regression is a supervised learning technique used to predict continuous numerical values by learning relationships between input variables (features) and an output variable (target).

[Linear Regression] is one of the simplest and most widely used regression algorithms.

Conditions for applying Linear Regression:
- The number of Input(s)/feature(s) has to be 1.
- The relationship between Input and Output must be linear $y = w_0 + w_1 x$

Examples of non-linear relationships:
- Polynomial: $y = w_0 + w_1 x + w_2 x^2$
- Exponential: $y = w_0 e^{w_1 x}$
- Logarithmic: $y = w_0 + w_1 \ln(x)$

[Logistic Regression] When the output is categorical, it can take only a limited number of possible values.

For example: Pass or Fail | Yes or No | Spam or Not Spam

In simple words, when the problem is a classification problem, we can use Logistic Regression.

For now, we assume two things:
- Only one input feature x is present.
- It is a binary classification problem; therefore, the output has two possible classes: y ∈ {0, 1}

Here, we assume only **one input feature** x and two possible output classes:

$$y \in \{0, 1\}$$

First, Logistic Regression calculates the linear value:

$$f = w_0 + w_1 x$$

This part will be given.

It then passes f through the **sigmoid function**:

$$z = \text{sigmoid}(f) = \frac{1}{1+e^{-f}}$$

The sigmoid output z is always between 0 and 1:

$$0 \le z \le 1$$

The sigmoid output z is converted into a class using a threshold.

$$\hat{y} = \begin{cases} 1, & \text{if } z \ge \text{threshold} \\ 0, & \text{if } z < \text{threshold} \end{cases}$$

Remember

z represents the probability that the input x belongs to class 1.

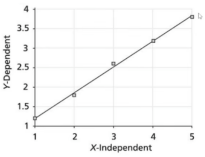
Thus, Logistic Regression converts:

$$\text{Input } x \longrightarrow f \longrightarrow z \longrightarrow \text{Class}$$

Linear Regression – Solved Example

- Let us consider an example where the five weeks' sales data (in Thousands) is given as shown in Table.
- Apply linear regression technique to predict the 7th and 12th week sales.

x_i (Week)	y_i (Sales in Thousands)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8



- Linear regression equation is given by
- $y = a_0 + a_1 * x + e$ (error)
- *where*
- $a_1 = \frac{(\bar{x}\bar{y}) - (\bar{x})(\bar{y})}{\bar{x^2} - \bar{x}^2}$
- $a_0 = \bar{y} - a_1 * \bar{x}$

- Here, there are 5 items, i.e., i = 1, 2, 3, 4, 5.

	x_i (Week)	y_i (Sales in Thousands)	x_i^2	$x_i * y_i$
	1	1.2	1	1.2
	2	1.8	4	3.6
	3	2.6	9	7.8
	4	3.2	16	12.8
	5	3.8	25	19
Sum	15	12.6	55	44.4
Average	$\bar{x} = 3$	$\bar{y} = 2.52$	$\bar{x^2} = 11$	$\bar{xy} = 8.88$

- $\bar{x} = 3$ $\bar{y} = 2.52$ $\bar{x^2} = 11$ $\bar{xy} = 8.88$
- $a_1 = \frac{(\bar{x}\bar{y}) - (\bar{x})(\bar{y})}{\bar{x^2} - \bar{x}^2} = \frac{8.88 - 3*2.52}{11 - 3^2} = 0.66$
- $a_0 = \bar{y} - a_1 * \bar{x} = 2.52 - 0.66 * 3 = 0.54$

- **Regression equation is**
- $y = a_0 + a_1 * x$
- $y = 0.54 + 0.66 * x$
- The predicted 7th week sale (when x = 7) is,
- **y = 0.54 + 0.66 x 7 = 5.16**
- the predicted 12th week sale (when x = 12) is,
- **y = 0.54 + 0.66 x 12 = 8.45**

Logistic Regression – Solved Example 1

- The dataset of pass or fail in an exam of 5 students is given in the table.
- Use logistic regression as classifier to answer the following questions.

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

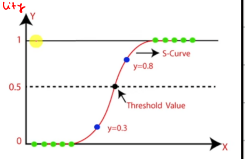
1. Calculate the probability of pass for the student who studied 33 hours.
2. At least how many hours student should study that makes he will pass the course with the probability of more than 95%.

Assume the model suggested by the optimizer for odds of passing the course is, (chance)

$$\log(odds) = -64 + 2 * hours$$

- We use Sigmoid Function in logistic regression (outputs [0→1], probability)

$$s(x) = \frac{1}{1+e^{-x}}$$



1. Calculate the probability of pass for the student who studied 33 hours.

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

- $p = \frac{1}{1+e^{-z}}$ $s(x) = \frac{1}{1+e^{-x}}$
- $z = -64 + 2 * 33 = -64 + 66 = 2$
- $p = \frac{1}{1+e^{-2}} = 0.88$
- That is, if student studies 33 hours, then there is **88% chance** that the student will pass the exam

$$\log(odds) = z = -64 + 2 * hours$$

2. At least how many hours student should study that makes he will pass the course with the probability of more than 95%.

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

- $p = \frac{1}{1+e^{-z}} = 0.95$
- $0.95 * (1 + e^{-z}) = 1$
- $0.95 * e^{-z} = 1 - 0.95$
- $e^{-z} = \frac{0.05}{0.95} = 0.0526$
- $\ln(e^{-z}) = \ln(0.0526)$

$\ln(e^x) = x$

$-z = \ln(0.0526) = -2.94$

$z = 2.94$

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

- $z = 2.94$
- $\log(odds) = z = -64 + 2 * hours$
- $2.94 = -64 + 2 * hours$
- $2 * hours = 2.94 + 64$
- $2 * hours = 66.94$
- $hours = \frac{66.94}{2}$
- $hours = 33.47\ Hours$

- The student should study **at least 33.47 hours**, so that he will pass the exam with more than 95% probability

Why can't we just use linear regression?

Imagine using a straight line to predict whether someone has a disease.

The output could become:

$$-0.7, \quad 0.3, \quad 0.8, \quad 1.5$$

But probability must be between:

$$0 \le P \le 1$$

So logistic regression uses a special **S-shaped function**, called the **sigmoid function**, to convert the linear result into a probability:

$$P(y=1) = \frac{1}{1+e^{-z}}$$

where

$$z = b_0 + b_1 x$$

So the process is basically:

$$x \rightarrow \text{linear equation} \rightarrow \text{sigmoid} \rightarrow \text{probability} \rightarrow \text{class}$$

Titanic Survival Problem

Suppose we have a simplified version of the Titanic dataset:

Passenger	Age (x)	Survived (y)
Passenger 1	22	False
⋮	⋮	⋮

Suppose Logistic Regression gives the following model:

$$f = 5 - 0.1x$$

Passing f through the sigmoid function gives:

$$z = \text{sigmoid}(f) = \frac{1}{1+e^{-f}}$$

Threshold $= 0.5$

answer the following questions:

1. For a passenger who is 55 years old, will the model predict that the passenger survives?
2. If a passenger has a 45% predicted chance of survival, approximately how old is the passenger?

(1) $f = 5 - 0.1x$; $z = \frac{1}{1+e^{-f}}$; 55 years old survive?

$f = 5 - 0.1 \times 55 = -0.5$

$\therefore z = \frac{1}{1+e^{-f}} = \frac{1}{1+e^{0.5}} = 0.378$

$z = 0.378 < 0.5$ [Doesn't survive]

(2) If 45% predicted chance of survival, how old is he?

$z = \frac{1}{1+e^{-f}} = 0.45$

$\Rightarrow 1 = 0.45\,(1+e^{-f})$

$\Rightarrow 1 = 0.45 + 0.45 \times e^{-f}$

$\Rightarrow 0.55 = 0.45\, e^{-f}$

$\Rightarrow e^{-f} = \frac{0.55}{0.45}$

$\Rightarrow \ln(e^{-f}) = \ln(\frac{0.55}{0.45})$

$\Rightarrow -f = 0.2$

$\therefore f = -0.2007$

$\therefore f = 5 - 0.1x$

~~$\Rightarrow 0.2 = 5 - 0.1x$~~

~~$\Rightarrow 0.1x = 5 - 0.2$~~

~~$\Rightarrow x = 48$ years~~

$\Rightarrow -0.2007 = 5 - 0.1x$

$\Rightarrow 0.1x = 5 + 0.2007$

$\Rightarrow x = 52.007$ years

We use logistic regression:

$$y = \frac{1}{1+e^{-z}}$$

where:

$$z = -27 + 3x$$

Here,

- x = temperature
- y = probability of playing

1. Find Probability at 33F?
2. Temperature of 91% probability?

26/31 $y = \frac{1}{1+e^{-z}}$; $z = -27 + 3x$ | x = temperature, y = P(playing)

(1) P(playing) at 33°F?

$\Rightarrow z = -27 + 3(33) = 72$

$\therefore y = \frac{1}{1+e^{-72}} = 1$

$\Rightarrow$ P(play) = 1

(2) temperature for 91% probability?

$\Rightarrow y = \frac{1}{1+e^{-z}}$

$\Rightarrow 0.91 = \frac{1}{1+e^{-z}}$

$\Rightarrow 0.91\,(1+e^{-z}) = 1$

$\Rightarrow 0.91 + 0.91\, e^{-z} = 1$

$\Rightarrow 0.91\, e^{-z} = 1 - 0.91$

$\Rightarrow \ln(e^{-z}) = \ln(\frac{1-0.91}{0.91})$

$\Rightarrow -z = -2.314$

$\therefore z = 2.314$

and, $z = -27 + 3x$

$\therefore 2.314 = -27 + 3x$

$\Rightarrow 3x = 2.314 + 27$

$\Rightarrow x = \frac{2.314+27}{3}$

$\Rightarrow x = 9.77°F$

Clustering(Unsupervised Learning)

Clustering is an unsupervised machine learning technique that groups similar data points together without being given predefined labels.

K-means clustering

Given data and a number k (number of clusters),
k-means groups the data into k clusters.

How K-Means Works -
1. Choose the number of clusters, k.
2. Randomly select k initial centroids.
3. For each data point, calculate the distance to all centroids.
4. Assign the point to the cluster whose centroid is closest(minimum distance)
5. For each cluster, recompute the centroid as the average of all points in that cluster.
6. Repeat steps 3-5 until the centroids stop changing (clusters become stable).

Result>> Similar data points end up in the same cluster.

Example> Classify the following data into 2 clusters using K-means.
(3,4), (6,7), (4,5), (5,7), (2,6)

Solution:

(i) Centroid selection

Let, Class-I : (2,6)

Class-II : (4,5)

Use euclidian distance

(2,6) (3,4)

$d = \sqrt{(2-3)^2 + (6-4)^2} =$

Iteration-1>

Data	Distance to		Cluster
	c_1-(2,6)	c_2-(4,5)	
(3,4)	2.236	1.414	c2
(6,7)	4.123	2.828	c2
(4,5)	2.236	0	c2
(5,7)	3.162	2.236	c2
(2,6)	0	2.236	c1

Iteration-2>

Data	Distance to		Cluster
	c1 (2,6)	c2 (4.5,5.75)	
(3,4)	2.236	2.305	c1
(6,7)	4.123	1.95	c2
(4,5)	2.236	0.90	c2
(5,7)	3.162	1.35	c2
(2,6)	0	-	c1

New centroids>

c2 ⇒ (3,4), (6,7), (4,5), (5,7)

$\Rightarrow \left(\frac{3+6+4+5}{4}, \frac{4+7+5+7}{4}\right)$

⇒ (4.5, 5.75)

c1 ⇒ (2,6)

New centroid>

c1 ⇒ (3,4), (2,6) = $\left(\frac{3+2}{2}, \frac{4+6}{2}\right) = (2.5, 5)$

c2 ⇒ (6,7), (4,5), (5,7) = (5, 6.33)

Iteration-3>

Data	Distance to		Cluster
	(2.5,5)	(5,6.33)	
(3,4)	1.12	3.07	c1
(6,7)	4.03	1.2	c2
(4,5)	1.5	1.66	c1
(5,7)	3.2	0.67	c2
(2,6)	1.12	3.01	c1

Iteration-4>

Data	Distance to		new cluster	Old cluster
	(3,5)	(5.5,7)		
(3,4)	1	3.5	c1	c1
(6,7)	3.6	0.5	c2	c2
(4,5)	1	2.5	c1	c1
(5,7)	2.8	0.5	c2	c2
(2,6)	1.4	3.6	c1	c1

As the cluster assignment remains same,
no further change occurs

Final Clusters are:
Cluster 1: (2,6), (3,4), (4,5) having centroid (3,5)
Cluster 2: (5,7), (6,7) having centroid (5.5, 7)

New centroids>

c1 ⇒ (3,4), (4,5), (2,6) = (3, 5)

c2 ⇒ (6,7), (5,7) = (5.5, 7)

classify this data (2,7) using the previous experiment.

Ans:

Class-I: (Distance) $d_1 = dist\{(2,7), (3,5)\}$

$= \sqrt{(2-3)^2 + (7-5)^2}$

$= \sqrt{1+2^2}$

$= \sqrt{5}$

$= 2.236$

Class-II: (Distance) $d_2 = dist\{(2,7), (5.5,7)\}$

$= \sqrt{(2-5.5)^2 + (7-7)^2}$

$= \sqrt{(-3.5)^2 + 0}$

$= \sqrt{12.25}$

$= 3.5$

Here 2.236 is smaller than 3.5

So the given data (2,7) is in class-I.

Consider the following dataset with 3 features and 5 examples:

ID	Age	Blood-Pressure	Blood-Sugar-Level
1	22	80	90
2	24	82	92
3	26	85	95
4	50	140	160
5	54	145	170

Task: Apply K-means clustering here with k = 2.

Solution:
Data points>> (22, 80, 90), (24, 82, 92), (26, 85, 95), (50, 140, 160), (54, 145, 170)

Initial centroids>> C1 = (22, 80, 90), C2 = (50, 140, 160)

Iteration 1: Distance Comparison

Current centroids:

$C_1^{(0)} = (22, 80, 90), \quad C_2^{(0)} = (50, 140, 160)$

Point	Distance to $C_1^{(0)}$	Distance to $C_2^{(0)}$	Cluster
(22, 80, 90)	0.00	96.95	C_1
(24, 82, 92)	3.46	93.49	C_1
(26, 85, 95)	8.12	87.92	C_1
(50, 140, 160)	96.95	0.00	C_2
(54, 145, 170)	108.02	11.18	C_2

So after the first assignment:

$C_1 = \{(22, 80, 90), (24, 82, 92), (26, 85, 95)\}$

$C_2 = \{(50, 140, 160), (54, 145, 170)\}$

new centroids by taking the mean of each cluster.
For Cluster 1: C1 = (22+24+26/3, 80+82+85/3, 90+92+95/3)
==> C1=(24,82.33,92.33)

For Cluster 2: C2 = (50+54/2, 140+145/2, 160+170/2)
==> C2 = (52, 142.5, 165)

Iteration 2: Distance Comparison

Now use the updated centroids:

$C_1^{(1)} = (24, 82.33, 92.33), \quad C_2^{(1)} = (52, 142.5, 165)$

Point	Distance to $C_1^{(1)}$	Distance to $C_2^{(1)}$	Cluster
(22, 80, 90)	3.28	100.75	C_1
(24, 82, 92)	0.47	97.44	C_1
(26, 85, 95)	4.11	91.78	C_1
(50, 140, 160)	91.89	5.59	C_2
(54, 145, 170)	102.96	5.59	C_2

From the distance table, the cluster assignment remains:
C1 = {(22, 80, 90), (24, 82, 92), (26, 85, 95)}
C2 = {(50, 140, 160), (54, 145, 170)}

So the points stay in the same clusters as before.

Since the cluster assignments remain the same,
the centroids also remain the same.
C1 = (24, 82.33, 92.33)
C2 = (52, 142.5, 165)

Therefore, no further change occurs.
Final clusters are:
Cluster 1: (22, 80, 90), (24, 82, 92), (26, 85, 95)
Cluster 2: (50, 140, 160), (54, 145, 170)

Modified K-Means Clustering Algorithm

Algorithm Steps>
Step 1: Select initial group centroids cleverly.
[Make a table and find distance of everyone, choose the two points with maximum distance.]
Step 2: Rest is same as general k-means algoritm

Example> Classify the following data points into k=2 clusters using the Modified K-Means Algoritms.
Data Points: (3,4), (6,7), (4,5), (5,7), (2,6)

Solution>

Step 1: Clever Initial Centroid Selection. To select initial centroids, construct a distance matrix calculating the Euclidean distance [d = sqrt{(x2-x1)^2 + (y2-y1)^2}] between every pair of data points.

Data Points	(3,4)	(6,7)	(4,5)	(5,7)	(2,6)
(3,4)	0	4.242	1.414	3.605	2.236
(6,7)	4.242	0	2.828	1.0	4.123
(4,5)	1.414	2.828	0	2.236	2.236
(5,7)	3.605	1.0	2.236	0	3.162
(2,6)	2.236	4.123	2.236	3.162	0

How the initial centroids Are Chosen?

Look across the table to find the points with the largest/maximum distance.
The maximum value in the table for distance is 4.242,
which corresponds to (6,7) and (3,4)
So:
Class-I Initial Centroid: (3,4)
Class-II Initial Centroid: (6,7)

**Rest of the part is same as normal K-means

Iteration 1>

Data Point	Distance to Class-I (3,4)	Distance to Class-II (6,7)	Assigned Class
(3,4)	0	4.242	Class-I
(6,7)	4.242	0	Class-II
(4,5)	1.414	2.828	Class-I
(5,7)	3.605	1.0	Class-II
(2,6)	2.236	4.123	Class-I

Cluster Groupings>>
Class-I: (3,4), (4,5), (2,6)
Class-II: (6,7), (5,7)

Recalculating Centroids(Mean)
Class-I Centroid: (3+4+2/3, 4+5+6/3) = (3,5)
Class-II Centroid: (6+5/2, 7+7/2) = (5.5,7)

Iteration 2>

Data Point	Distance to Class-I (3,5)	Distance to Class-II (5.5,7)	Assigned Class
(3,4)	1.0	3.91	Class-I
(6,7)	3.61	0.5	Class-II
(4,5)	1.0	2.5	Class-I
(5,7)	2.83	0.5	Class-II
(2,6)	1.414	3.64	Class-I

Because the cluster assignments remained unchanged between Iteration 1 and Iteration 2, the algorithm stops.

Final Output Table

Data Points	Class	Final Centroid
$(3,4),(4,5),(2,6)$	Class-I	(3, 5)
$(6,7),(5,7)$	Class-II	(5.5, 7)

Neural Network(Supervised Learning)

Term	What does it mean?	Key idea
Neural network	A model made of interconnected neurons	Weights and biases
Deep learning	Learning with neural networks having multiple hidden layers	Multiple layers
Supervised learning	Learning from inputs with known correct outputs	Labeled data

What is an Artificial Neural Network (ANN)?
An Artificial Neural Network is a machine learning model inspired by the structure and functioning of the human brain. It consists of interconnected artificial neurons that learn patterns from data by adjusting weights and biases.

Q2. What is an artificial neuron?

An artificial neuron is the basic processing unit of a neural network. It receives inputs, multiplies them by weights, adds a bias, and applies an activation function to produce an output.

Its mathematical form is:

$$z = \sum_{i=1}^{n} w_i x_i + b$$

$$y = f(z)$$

Here:

- x_i: input values
- w_i: weights representing the importance of inputs
- b: bias, which shifts the neuron's decision boundary
- z: weighted sum plus bias
- $f(z)$: activation function's output

What is Perceptron?
A perceptron is a simple model of a biological neuron that is used in Artificial Neural Networks. It is a machine based algorithm that learns how to classify data.

What is the difference between a neuron and a perceptron?
A perceptron is a specific, simple type of artificial neuron, usually using a step activation function. The term artificial neuron is broader and includes neurons using functions such as sigmoid.

Q5. What are weights and why are they used?
Weights are trainable numerical parameters that determine how strongly each input influences the neuron's output.
- A large positive weight increases the influence in the positive direction.
- A negative weight influences the output in the opposite direction.
- A weight close to zero means that the input has relatively little influence.

Weights are adjusted during training to improve predictions.

What is bias in a neural network?
Bias is a trainable parameter added to the weighted sum. It allows the neuron to shift its decision boundary independently of the input values.

What is an Activation function?
An activation function decides whether a neuron should be activated or not. Means it'll decide whether the neuron's input to the network is important or not in the process of prediciton using simple mathematical operations.

1. Step function

$$f(z) = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

Produces a binary output, usually 0 or 1. It is commonly associated with the basic perceptron.

2. Sigmoid function

$$f(z) = \frac{1}{1 + e^{-z}}$$

Produces values between 0 and 1. It is useful for binary classification outputs interpreted as probabilities.

4. How does a perceptron learn?

You already know the formulas. Now understand the purpose of each step.

Step 1: Forward propagation
Inputs move through the network to produce a prediction.
Why? The network needs to calculate its current answer.

Step 2: Calculate the weighted sum
Multiply each input by its weight, then add the bias.
Why? This combines the input information into a single value.

Step 3: Calculate the error
Compare the predicted output with the actual target.
Why? The error tells us how the prediction differs from the desired answer.

Step 4: Update weights and bias
Adjust the parameters based on the error and learning rate.
Why? The network needs to change its future predictions to improve learning.

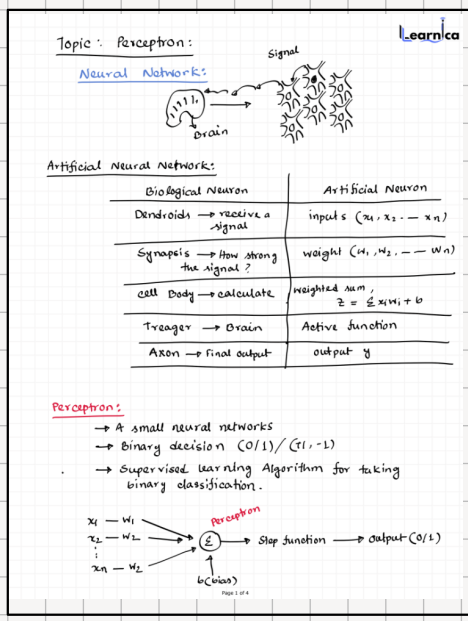
Step 5: Repeat training
Process more training examples and update the parameters as needed.
Why? Repeated learning helps the model discover a better decision boundary.

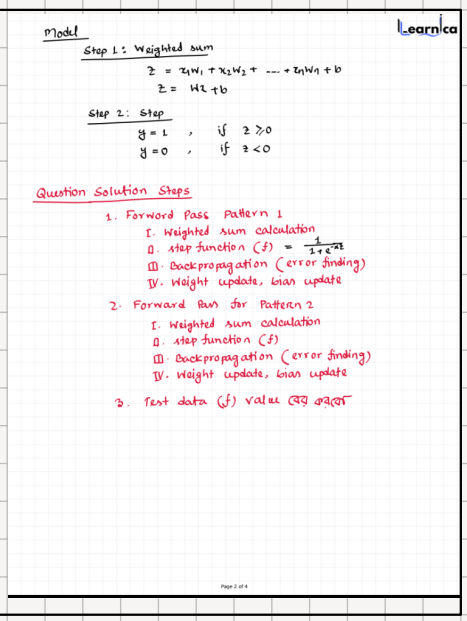
What is forward propagation?
Forward propagation is the process of passing input data through the network, calculating weighted sums and activations, to obtain the predicted output.

What is an error in a neural network?
Error is the difference between the target output and the predicted output. It indicates how far the prediction is from the desired answer.

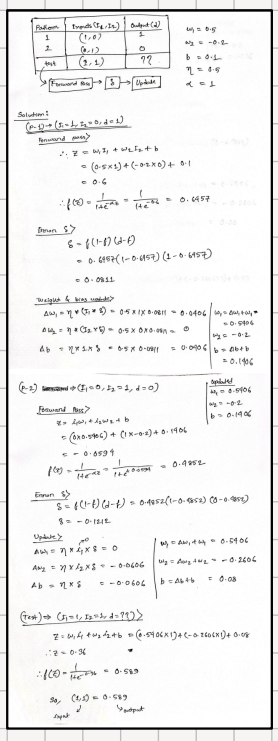
What is the learning rate?
The learning rate is a hyperparameter that controls the size of parameter updates during training.
- Too high: training may overshoot or become unstable.
- Too low: training may take a long time.

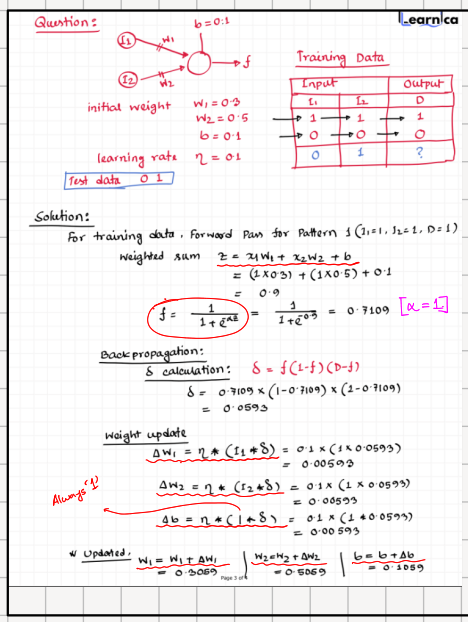
What is weight updating?
Weight updating is the process of modifying weights based on the error to improve future predictions.

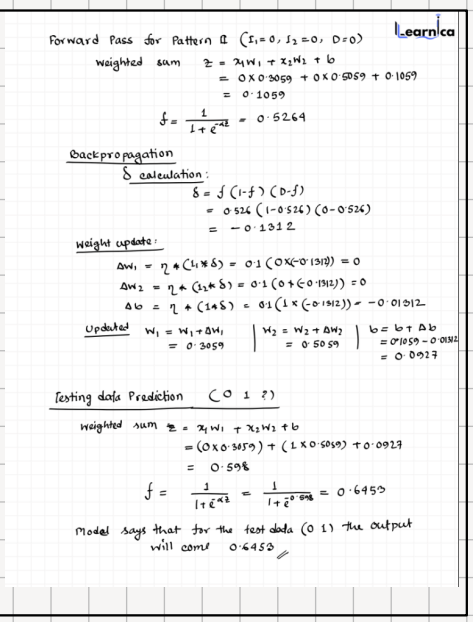




1. **Weighted Sum:** $z = \sum x_i w_i + b$
   PNG
2. **Activation:** $f(z) = \frac{1}{1+e^{-\alpha z}} \quad [\alpha = 1]$
   PNG
3. **Error:** $\delta = f(1-f)(d-f)$
   PNG
4. **Change:**
   - $\Delta w_i = \eta \times (I \times \delta)$
     PNG
   - $\Delta b = \eta \times (1 \times \delta)$
     PNG
5. **Update:** $w_i = \Delta w_i + w_i$
   PNG







Monte Carlo algorithm

A Monte Carlo algorithm is an algorithm for computers.
It is used to simulate the behavior of systems involving uncertainty, to predict and understand the behaviour of the system.

Monte Carlo

* Simulation: is the process of designing a model of a real system and conducting experiments with the model for the purpose of understanding the behaviour for the operation of the system.
  - → A duplication of an original system.
  - → To understand the implementation/out put of the sys-.

* Monte Carlo simulation technique:
  - → It is an experiment on probability
  - → Uses random numbers & requires decision-making under uncertainties.

Steps:
- (S-1) establishing probability distribution
- (S-2) cumulative "
- (S-3) Setting random num number intervals
- (S-4) Generating random numbers
- (S-5) to find the answer of question asked using the above four steps.

*Calculating Probability

Suppose we perform 10 simulations and event A occurs 6 times.

Then:

$$I_1+I_2+\dots\dots\dots\dots\dots+I_{10}=6$$

Therefore,

$$\hat{P}(A) = \frac{6}{10} = 0.6$$

Why does Monte Carlo work?

The mathematical foundation is the **Law of Large Numbers**.

Average $\frac{1}{N}\sum_{i=1}^{N} X_i \to E[X]$ expected value of x

If N→∞

Thus,

$$\hat{\mu} \to \mu$$

where $\mu = \int x f(x) dx$

This means that **more simulations generally produce a more accurate estimate**.

Monte Carlo error

The standard error of the Monte Carlo estimate is approximately

$$SE = \frac{\sigma}{\sqrt{N}}$$

where:
- σ = standard deviation of the population
- N = number of simulations

This gives an important relationship:

$$Error \propto \frac{1}{\sqrt{N}}$$

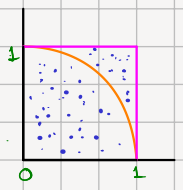
Simulations	Relative error
100	=1/10
1,000	=1/31.6
10,000	= 1/100
100,000	= 1/316

So increasing simulations by **100 times** reduces the typical Monte Carlo error by about **10 times**.

Monte Carlo Tree Search (MCTS): Monte Carlo does not calculate exact value. It estimates by repeatedly simulating possible outcomes and averaging their returns.

*Estimating the value of Pi(π)

Consider a square: [0≤x≤1], [0≤y≤1]
Inside it, consider a quarter circle: [x^2 + y^2 ≤ 1]



$A_{square} = 1 \times 1 = 1$

$A_{circle} = \frac{\pi r^2}{4} = \frac{\pi(1)^2}{4} = \frac{\pi}{4}$

So, $\frac{A_c}{A_s} = \frac{\pi/4}{1}$

$\Rightarrow \pi = \frac{A_c}{A_s} \times 4$

Now generate N random points [(xi,yi)~ U(0,1)] and distribute uniformly inside the grid [0->1], so it doesnt fall outside our desired range 0 to 1 in the grid

Count the points:
Suppose, N = total number of random points = 10,000
Nc= number of points inside the quarter-circle = 7,850

Then the estimated area ratio is -

$$\frac{A_c}{A_s} = \frac{N_c}{N} = \frac{7850}{10,000} = 0.785$$

Therefore $\pi = \frac{A_c}{A_s} \times 4$

$= 0.785 \times 4$

$= 3.14$

**This will only work if the random points generated falls uniformly inside the grid.

Problem Statement

Dr. Strong is a dentist who schedules all her patients for 30-minute appointments. Some patients take more or less than 30 minutes depending on the type of dental work to be done. The following summary shows the various categories of work, their probabilities, and the time actually needed to complete the work: JPG + 2

Category	Time Required	No. of Patients	Probability
Filling	45 min	40	0.40 = 40/100
Crown	60 min	15	0.15 = 15/100
Cleaning	15 min	15	0.15 = 15/100
Extracting	45 min	10	0.10 = 10/100
Checkup	15 min	20	0.20 = 20/100
		↳ = 100	

Task Requirements

1. **Simulate** the dentist's clinic for 4 hours starting at 8:00 a.m. JPG
2. Assume all patients show up at the clinic at exactly their scheduled arrival time (every 30 minutes). JPG
3. Find out the **average waiting time** for the patients as well as the **idleness of the doctor**. JPG
4. **Random Numbers to Use:** `40, 82, 11, 34, 25, 66, 17, 79.` JPG

Solution

Table I: Random Number Allocation Table

Intervals

Category	Probability	Cumulative Probability	Random Number Range
Filling	0.40	0.40	00 - 39
Crown	0.15	0.55	40 - 54 → crown
Cleaning	0.15	0.70	55 - 69
Extracting	0.10	0.80	70 - 79
Checkup	0.20	1.00	80 - 99

Table II: Patient Work Determination

Note: 4 hours of clinic time with 30-minute slots yields $4 \times 2 = 8$ scheduled patients starting at 8:00 a.m.

Patient	Scheduled Arrival	Random Number	Category	Service Time Needed
1	8:00	40	Crown	60 min
2	8:30	82	Checkup	15 min
3	9:00	11	Filling	45 min
4	9:30	34	Filling	45 min
5	10:00	25	Filling	45 min
6	10:30	66	Cleaning	15 min
7	11:00	17	Filling	45 min
8	11:30	79	Extracting	45 min

Table III: Simulation Table

Patient	Arrival	Service Start	Service Duration	Service Ends	Waiting Time (mins)	Idle Time (mins)
1	8:00	8:00	60 min	9:00	0	0
2	8:30	9:00	15 min	9:15	30	0
3	9:00	9:15	45 min	10:00	15	0
4	9:30	10:00	45 min	10:45	30	0
5	10:00	10:45	45 min	11:30	45	0
6	10:30	11:30	15 min	11:45	60	0
7	11:00	11:45	45 min	12:30	45	0
8	11:30	12:30	45 min	1:15	60	0
Total					285	0

Final Calculations

- $$\text{Average Waiting Time} = \frac{\text{Total Waiting Time}}{\text{Total Patients}} = \frac{285}{8} = 35.625 \text{ minutes}$$ JPG
- Total Doctor Idle Time = **0** minutes

Problem: Ramesh is a dentist who scheduled all his patients for 30 min appointments. Some patients take more or less than 30 min depending on the type of dental work to be done. The following summary shows various categories of work, their probability, and the time actually needed to complete work. Simulate the dentist's clinic for 4 hrs and find out the average waiting time for the patient as well as the ideal time of doctor. Assume that all patient reaches at clinic exactly as per schedule.

Category	Time Required (min)	No. of Patients	Probability
Filling	45	30	0.30
Crowning	60	15	0.15
Cleaning	15	10	0.10
Extracting	45	25	0.25
Checkup	15	20	0.20
Total	—	**100**	**1.00**

Using Random numbers: - 42, 82, 10, 34, 25, 66, 17, 79

"No. of patient visiting within 4hrs are 8 patient."

using Random numbers :- 42, 82, 10, 34, 25, 66, 17, 79

"No : of patient visiting within 4hrs are 8 patient."

Category	Probability	Cumm. Probability	Random No. of Interval
Filling	0.3	0.3	0 - 29
Crowning	0.15	0.45	30 - 44
Cleaning	0.10	0.55	45 - 54
Extracting	0.25	0.8	55 - 79
Checkup	0.2	1.0	80 - 99

Patient	Schedule Arrival	Random No.	Category	Service Time Needed
1	8:00	42	Crowning	60
2	8:30	82	Check up	15
3	9:00	10	Filling	45
4	9:30	34	Crowning	60
5	10:00	25	Filling	45
6	10:30	66	Extracting	45
7	11:00	17	Filling	45
8	11:30	79	Extracting	45

Patient	Arrival	Service Start	Service Duration	Service End	Waiting Time (min)	Idle time For doctor
1	8:00	8:00	60	9:00	0	0
2	8:30	9:00	15	9:15	30	0
3	9:00	9:15	45	10:00	15	0
4	9:30	10:00	60	11:00	30	0
5	10:00	11:00	45	11:45	60	0
6	10:30	11:45	45	12:30	75	0
7	11:00	12:30	45	1:15	90	0
8	11:30	1:15	45	2:00	105	0

$\Sigma t = 405$ min

$$\text{avg} = \frac{\Sigma t}{n} = \frac{405}{8} = 50.625 \text{ min}$$

→ We can make the solution in 2-tables

Table-1

Category	Probability	Cumm. Probability	Random No. of Interval
Filling	0.3	0.3	0 - 29
Crowning	0.15	0.45	30 - 44
Cleaning	0.10	0.55	45 - 54
Extracting	0.25	0.8	55 - 79
Checkup	0.2	1.0	80 - 99

Table-2

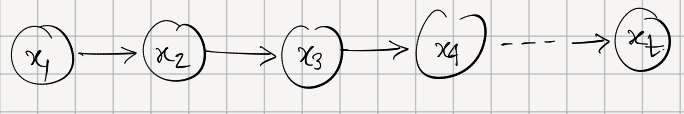
Patient	Arrival	Start	End	Service time	Random Number	Category	Waiting	Idle
1	8:00	8:00	9:00	60	42	Crowning	0	0
2	8:30	9:00	9:15	15	82	Checkup	30	0
3	9:00	9:15	10:00	45	10	Filling	15	0
4	9:30	10:00	11:00	60	34	Crowning	30	0
5	10:00	11:00	11:45	45	25	Filling	60	0
6	10:30	11:45	12:30	45	66	Extracting	75	0
7	11:00	12:30	01:15	45	17	Filling	90	0
8	11:30	01:15	02:00	45	79	Extracting	105	0
							405	0

→ Average waiting = $\frac{405}{8}$ = 50.625

→ Doctor's idle = 0

Markov Models

A Markov model is a mathematical model used to predict what might happen next based on only the current state of a system.



$$P(X_t) = P(X_t \mid X_{t-1})$$

Stationarity assumption: transition probabilities stays same at all times.
Current state only depends on the last state

- States: X = {rain, sun}
- Initial distribution: 1.0 sun ($P(x_1=\text{sun}) = 1.0$, $P(x_1=\text{rain}) = 0.0$)
- CPT $P(X_t \mid X_{t-1})$:

X_{t-1}	X_t	$P(X_t \mid X_{t-1})$
sun	sun	0.9
sun	rain	0.1
rain	sun	0.3
rain	rain	0.7

$P(x_2=\text{sun}) = ?$
$P(x_2=\text{rain}) = ?$

$P(x_3=\text{sun}) = ?$
$P(x_3=\text{rain}) = ?$

$P(x|y) = \frac{P(x,y)}{P(y)} \Rightarrow P(x,y) = P(x|y) \cdot P(y)$

Solution>

$$P(x_2=\text{sun}) = P(x_2=s, x_1=s) + P(x_2=s, x_1=r)$$
$$= P(x_2=s \mid x_1=s) \cdot P(x_1=s) + P(x_2=s \mid x_1=r) \cdot P(x_1=r)$$
$$= (0.9 \times 1) + (0.3 \times 0)$$
$$= 0.9$$

(1 = given initial probability)

$$\therefore P(x_2=\text{rain}) = 1 - P(x_2=\text{sun})$$
$$= 1 - 0.9$$
$$= 0.1$$

So, $P(x_2=\text{sun}) = 0.9$
$P(x_2=\text{rain}) = 0.1$

$$\therefore P(x_3=\text{sun}) = P(x_3=s, x_2=s) + P(x_3=s, x_2=r)$$
$$= P(x_3=s \mid x_2=s)\, P(x_2=s) + P(x_3=s \mid x_2=r)\, P(x_2=r)$$
$$= (0.9 \times 0.9) + (0.3 \times 0.1)$$
$$= 0.84$$

$$\therefore P(x_3=\text{rain}) = 1 - P(x_3=\text{sun})$$
$$= 1 - 0.84$$
$$= 0.16$$

#Calculate at t=infinity

$P_{\alpha}(x) = P_{\alpha+1}(x) \cdots$ (stationary distribution)

CPT $P(X_t \mid X_{t-1})$:

X_{t-1}	X_t	$P(X_t \mid X_{t-1})$
sun	sun	0.9
sun	rain	0.1
rain	sun	0.3
rain	rain	0.7

$$P_{\alpha}(\text{sun}) = P_{\alpha+1}(\text{sun} \mid \text{sun})\, P_{\alpha}(\text{sun}) + P_{\alpha+1}(\text{sun} \mid \text{rain})\, P_{\alpha}(\text{rain})$$
$$\Rightarrow a = P_{\alpha}(\text{sun} \mid \text{sun})\, P_{\alpha}(\text{sun}) + P_{\alpha}(\text{sun} \mid \text{rain})\, P_{\alpha}(\text{rain})$$
$$\Rightarrow a = (0.9 \times a) + (0.3 \times b)$$
$$\Rightarrow a = 0.9a + 0.3b \cdots (i)$$

Similarly

$$P_{\alpha}(\text{rain}) = P_{\alpha}(\text{rain} \mid \text{rain}) \cdot P_{\alpha}(\text{rain}) + P_{\alpha}(\text{rain} \mid \text{sun})\, P_{\alpha}(\text{sun})$$
$$\Rightarrow b = (0.7 \times b) + (0.1 \times a)$$
$$\Rightarrow b = 0.7b + 0.1a \cdots (ii)$$

↳ We could stop this and do

$$P_{\alpha}(\text{sun}) + P_{\alpha}(\text{rain}) = 1$$
$$\Rightarrow a + b = 1 \cdots (iii)$$
$$\hookrightarrow a = 1 - b \cdots (iv)$$

Put (a) in (i)

$$a = 0.9a + 0.3b$$
$$\Rightarrow (1-b) = 0.9(1-b) + 0.3b$$
$$\Rightarrow 1 - b = 0.9 - 0.9b + 0.3b$$
$$\Rightarrow 1 - 0.9 = b - 0.9b + 0.3b$$
$$\Rightarrow 0.1 = 0.4b$$
$$\therefore b = 0.25$$

Put 'b' in (iv)

$$a = 1 - b$$
$$\Rightarrow a = 1 - 0.25 = 0.75$$

$$\therefore a = P_{\alpha}(\text{sun}) = 0.75$$
$$\therefore b = P_{\alpha}(\text{rain}) = 0.25$$

- From initial observation of sun

$\begin{pmatrix}1.0\\0.0\end{pmatrix}_{P(X_1)}$ $\begin{pmatrix}0.9\\0.1\end{pmatrix}_{P(X_2)}$ $\begin{pmatrix}0.84\\0.16\end{pmatrix}_{P(X_3)}$ $\begin{pmatrix}0.804\\0.196\end{pmatrix}_{P(X_4)}$ $\Rightarrow \begin{pmatrix}0.75\\0.25\end{pmatrix}_{P(X_\infty)}$

- From initial observation of rain

$\begin{pmatrix}0.0\\1.0\end{pmatrix}_{P(X_1)}$ $\begin{pmatrix}0.3\\0.7\end{pmatrix}_{P(X_2)}$ $\begin{pmatrix}0.48\\0.52\end{pmatrix}_{P(X_3)}$ $\begin{pmatrix}0.588\\0.412\end{pmatrix}_{P(X_4)}$ $\Rightarrow \begin{pmatrix}0.75\\0.25\end{pmatrix}_{P(X_\infty)}$

- From yet another initial distribution $P(X_1)$:

$\begin{pmatrix}p\\1-p\end{pmatrix}_{P(X_1)}$ ... $\Rightarrow \begin{pmatrix}0.75\\0.25\end{pmatrix}_{P(X_\infty)}$

↳ Running 'x' simulations will give same result for every case. This is stationary distribution $[P_{\alpha}]$ of this chain.

$$P_{\alpha}(x) = P_{\alpha+1}(x)$$

{Sun, rain, overcast}

$P(x_{t-1})$	$P(x)$	values
sun	sun	–
sun	rain	–
sun	over	–
rain	sun	–
rain	rain	–
rain	over	–
over	sun	–
over	rain	–
over	over	–

$$P(x_2=\text{sun}) = P(x_2=s \mid x_1=s) \cdot P(x_1=s) +$$
$$P(x_2=s \mid x_1=r) \cdot P(x_1=r) +$$
$$P(x_2=s \mid x_1=o) \cdot P(x_1=o)$$

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5. A music analytics company is studying listening patterns on **Spotify**, specifically tracking how users switch their primary artist preference among three major rap artists: **Eminem**, **Travis Scott**, and **Drake**. They collected data on users' most-played artists in **January 2023** and tracked how their preferences changed by **January 2024**.

- **Initial (January 2023)** survey results **(total of 180 users)**:
  - **80** primarily listened to **Eminem**
  - **60** primarily listened to **Travis Scott**
  - **40** primarily listened to **Drake**

- One year later (**January 2024**), the following preference shifts were observed:
  - Among the **80 Eminem** listeners in **2023**:
    - **16** switched to **Travis Scott** as their primary artist
    - **12** switched to **Drake**
    - The rest stayed loyal to **Eminem**
  - Among the **60 Travis Scott** listeners in **2023**:
    - **15** switched to **Eminem**
    - **9** switched to **Drake**
    - The rest continued with **Travis Scott** as their top artist
  - Among the **40 Drake** listeners in **2023**:
    - **10** switched to **Eminem**
    - **6** switched to **Travis Scott**
    - The rest kept **Drake** as their most-played artist

Assume these **transition patterns** remain the same for each subsequent year (**2024→2025, 2025→2026, 2026→2027 etc.**).

Questions

1. Construct a **transition matrix** representing the listener preference changes from any year **t** to the next year **t+1**. **[2]**
2. Predict the percentage of users who will have **Travis Scott** as their primary artist in **2028** using the initial survey of **2023**. **[3]**
3. Calculate the **long-run probabilities** that a randomly selected Spotify user will have for each of the three artists. **[3]**

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Initially (2023)

Eminem: $\frac{80}{180} = 0.4444$ | Travis: $\frac{60}{180} = 0.3333$ | Drake: $\frac{40}{180} = 0.2222$

So, initially

$V_0 = \begin{bmatrix} (E) & (T) & (D) \\ 0.4444 & 0.3333 & 0.2222 \end{bmatrix}$

In 2024

From (E), (80)

To (T) $= \frac{16}{80} = 0.2$

To (D) $= \frac{12}{80} = 0.15$

stays $= \frac{80-16-12}{80} = 0.65$

From (T), (60)

To (E) $= \frac{15}{60} = 0.25$

To (D) $= \frac{9}{60} = 0.15$

Stays $= \frac{60-15-9}{60} = 0.6$

From (D), (40)

To (E) $= \frac{10}{40} = 0.25$

To (T) $= \frac{6}{40} = 0.15$

stays $= \frac{40-10-6}{40} = 0.6$

(81)

$P(X_t)$

	T	D	E
T	0.6	0.15	0.25
D	0.15	0.6	0.25
E	0.2	0.15	0.65

$P(X_{t-1})$

(Q2) With the initial survey of 2023, we can determine the survey output of 2028 directly with transition matrix

$\therefore V_n = V_0 \times P^n$ [2023 → 2028] ↳ 5 years gap

$\therefore V = \begin{bmatrix} (T) & (D) & (E) \\ 0.3333 & 0.2222 & 0.4444 \end{bmatrix} \times \begin{bmatrix} 0.6 & 0.15 & 0.25 \\ 0.15 & 0.6 & 0.25 \\ 0.2 & 0.15 & 0.65 \end{bmatrix}^5$

$\therefore V_{2028} = \begin{bmatrix} (T) & (D) & (E) \\ 0.31 & 0.27 & 0.42 \end{bmatrix}$

In 2028, 31% people will have Travis Scott as their primary artist.

same way, if we wanted to find 2030

$V_{2030} = V_0 \times P^7$ and so on.

(Q3) Long-term probability / $P_\infty(x)$

$P(X_t)$ | $P(X_t | X_{t-1})$ | $P(X_{t-1})$

$P = \begin{bmatrix} (T) & (D) & (E) \\ 0.6 & 0.15 & 0.25 \\ 0.15 & 0.6 & 0.25 \\ 0.2 & 0.15 & 0.65 \end{bmatrix} \begin{matrix} (T) \\ (D) \\ (E) \end{matrix}$

$\therefore P_\infty(T) = P_\infty(T|T) \cdot P_\infty(T) + P_\infty(T|D) \cdot P_\infty(D) + P_\infty(T|E) \cdot P_\infty(E)$

$\Rightarrow a = (0.6 \times a) + (0.15 \times b) + (0.2 \times c)$

$\Rightarrow 0.4a - 0.15b - 0.2c = 0 \quad \cdots (i)$

$\therefore P_\infty(D) = P_\infty(D|D) P_\infty(D) + P_\infty(D|T) P_\infty(T) + P_\infty(D|E) P_\infty(E)$

$\Rightarrow b = (0.6 \times b) + (0.15 \times a) + (0.15 \times c)$

$\Rightarrow -0.15a + 0.4b - 0.15c = 0 \quad \cdots (ii)$

$\therefore a + b + c = 1 \quad \cdots (iii)$

Put i, ii, iii in simultaneous Eqn

$a = P(T) = 0.31 \rightarrow 31\%$.

$b = P(D) = 0.27 \rightarrow 27\%$.

$c = P(E) = 0.42 \rightarrow 42\%$.